

Thread - 2026-02-18

<https://x.com/RiikonenMarko/status/2023930223054459330>

1/21 - 2026-02-18 01:19 UTC

17/18 Feb 2026, Rovaniemi. The highlight of the night was visually conspicuous 9° halo. There were three swarms in Sierijärvi right next to the snow gun. This one was the first, already going when I arrived. Car thermometer -14 C, must be among the warmest odd radii diamond dusts.

Original: <https://t.co/U07yQocSrW>

[video: 2023930223054459330-AQgVbKFkMXwAtySt.mp4]

2/21 - 2026-02-18 16:23 UTC

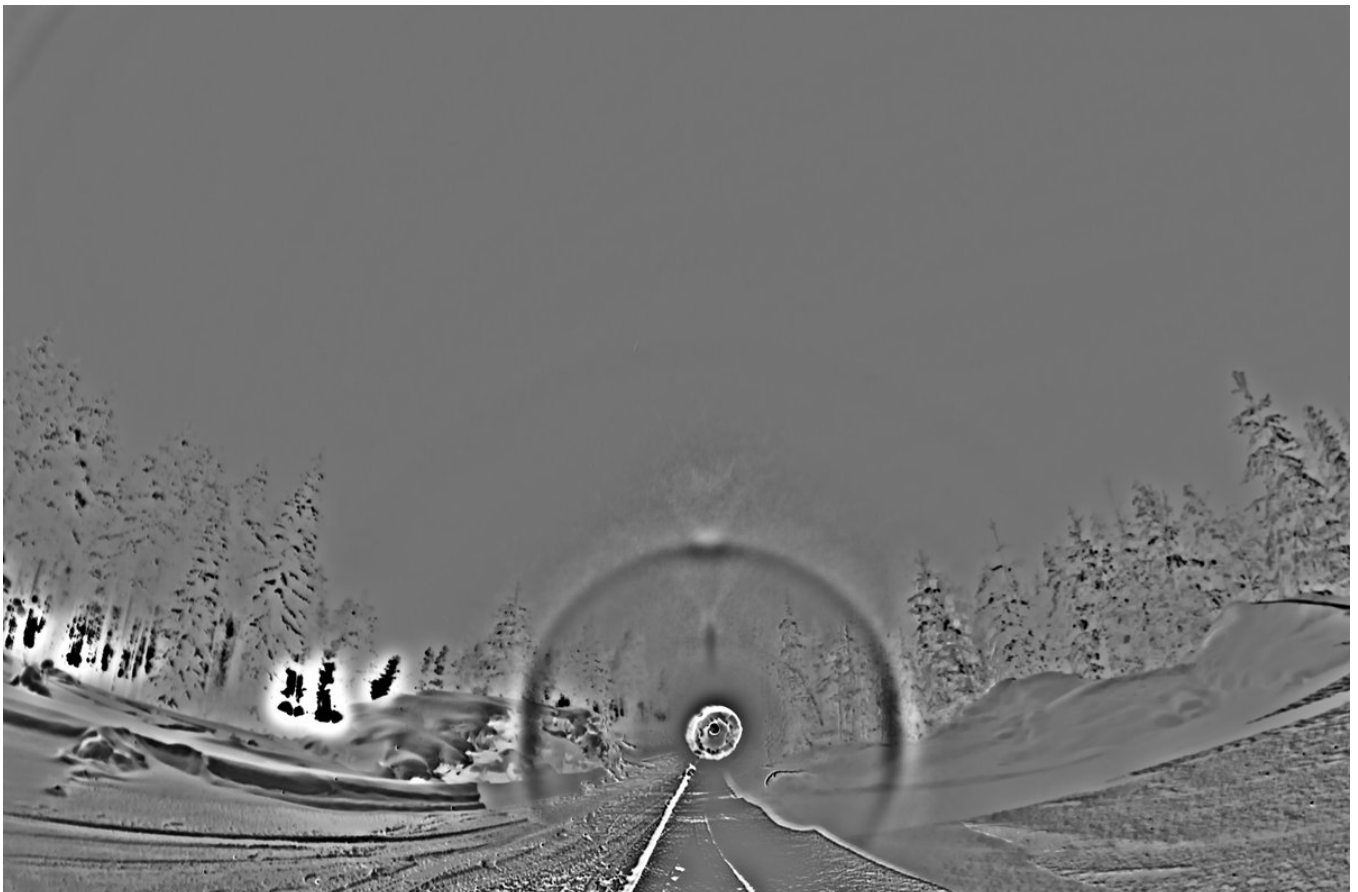
17/18 Feb 2026, Rovaniemi. The first swarm didn't have odd radii at the beginning. This stack of 77 first frames (10s exp) contains also frames where odd radii started weakly to manifest towards the end.

But what's the feature above the tanarc? It is reminiscent of what Mikkilä and Luomanen got in Kuusamo on 4 Dec 2023: <https://t.co/SNgUuuV43m> Now I am looking also differently at the "random shape from too little data" in my observation on 5/6 Feb 2026: <https://t.co/RddZsryvsV>

I need to work different length stacks to see if the arc gets improved. Also max.

What about that apparent red halo outside 22° halo in g-r and b-r? Just an artefact?

Lamp was on car roof, a little bit lower than camera because of the sloping ground. Maybe about -1 degs.







3/21 - 2026-02-18 17:01 UTC

17/18 Feb 2026, Rovaniemi. Before looking closer at the previous post's stuff, here is what followed when the swarm turned odd radii in earnest (50 x 10s). Too bad the location was so light polluted. No other choices for places because this stuff occurred strictly in the vicinity of the gun. There was no cloud whatsoever growing from the gunning.

On the previous night it had warmed to -5 C, windy and snowing. But forecast looked quite good for the evening, dropping temps and calm, so I left after the snowfall has stopped. Suosiola plume was healthy, continuing far in the distance. Like on the previous nights, I headed south, but it turned out that the backyard machine had fallen silent. Possibly they have now all the snow they need, or just had not yet restarted after the warm night. I thought of going back to apartment, but decided continue to Sierijärvi to see if Juujärvi racing has gotten their operation running despite all. And they had!





4/21 - 2026-02-20 20:17 UTC

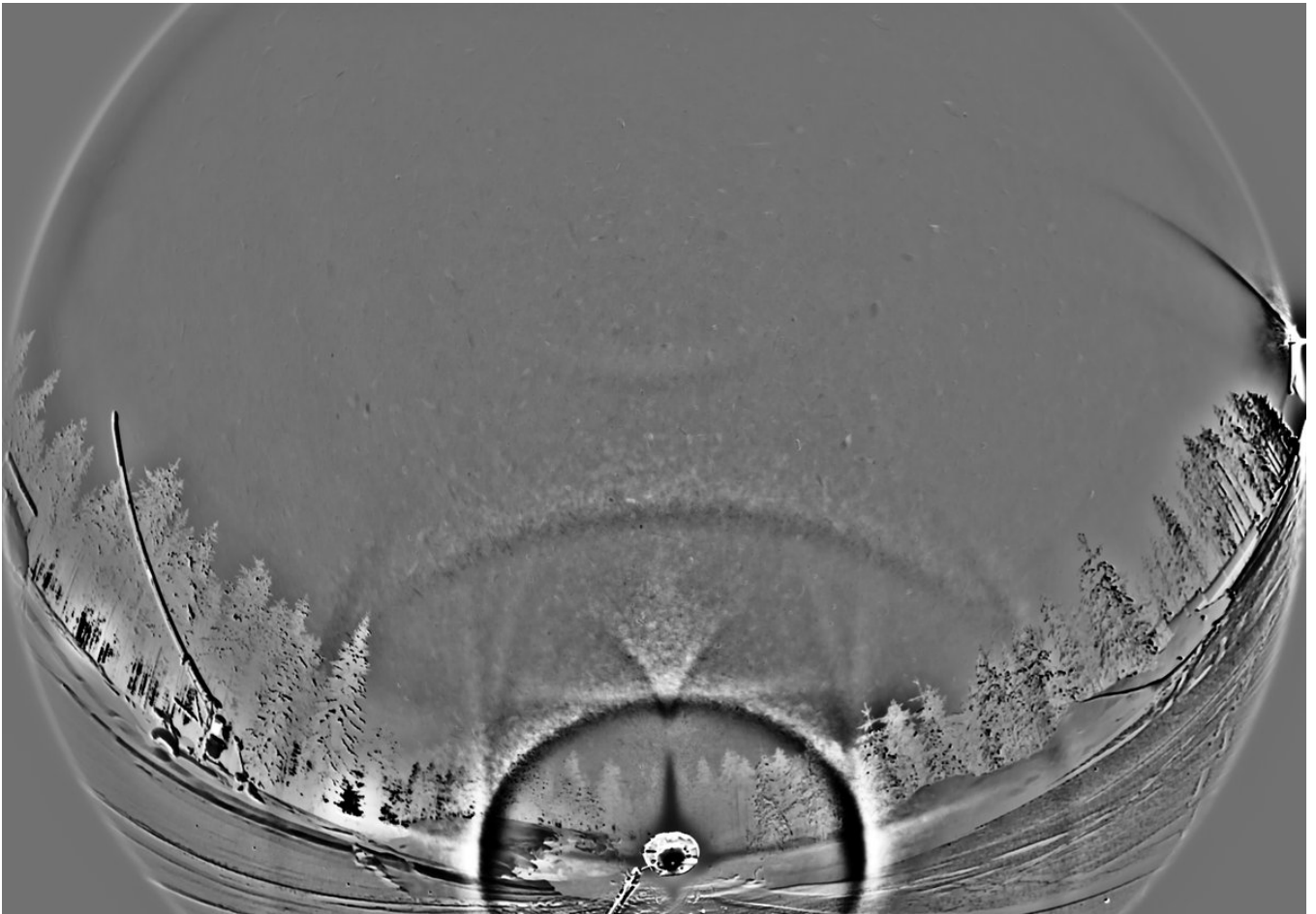
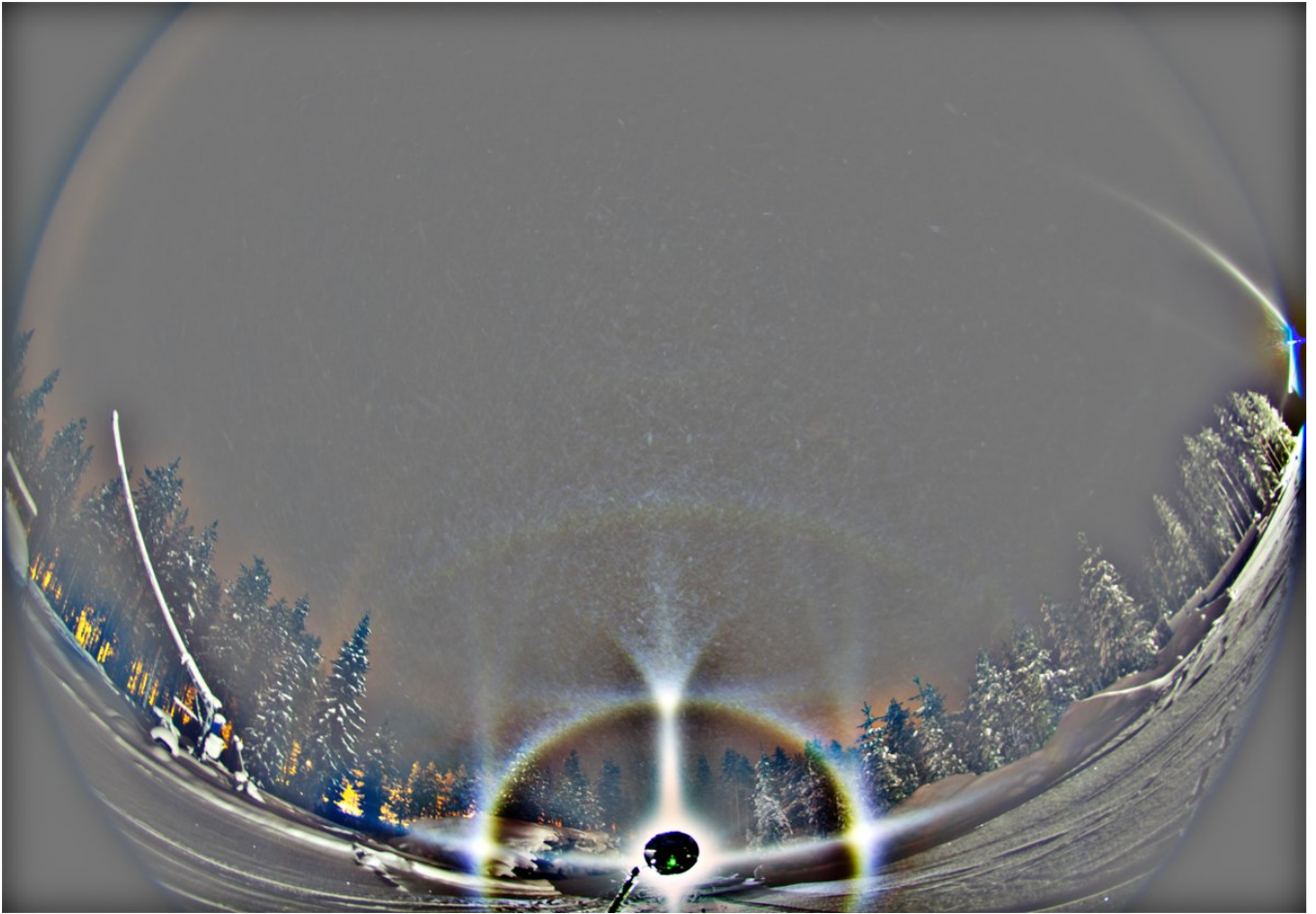
17/18 Feb 2026, Rovaniemi. After odd radius, the swarm disappeared. Then, at a snap of a fingers,

there was superparhelia on the building flood and I got busy to set up the spotlight again.



5/21 - 2026-02-20 20:23 UTC

17/18 Feb 2026, Rovaniemi. This is how it looked once I got the spotlight going. 13 x 30s. Hastings/Wegener is there and possibly also the "pseudo" version.



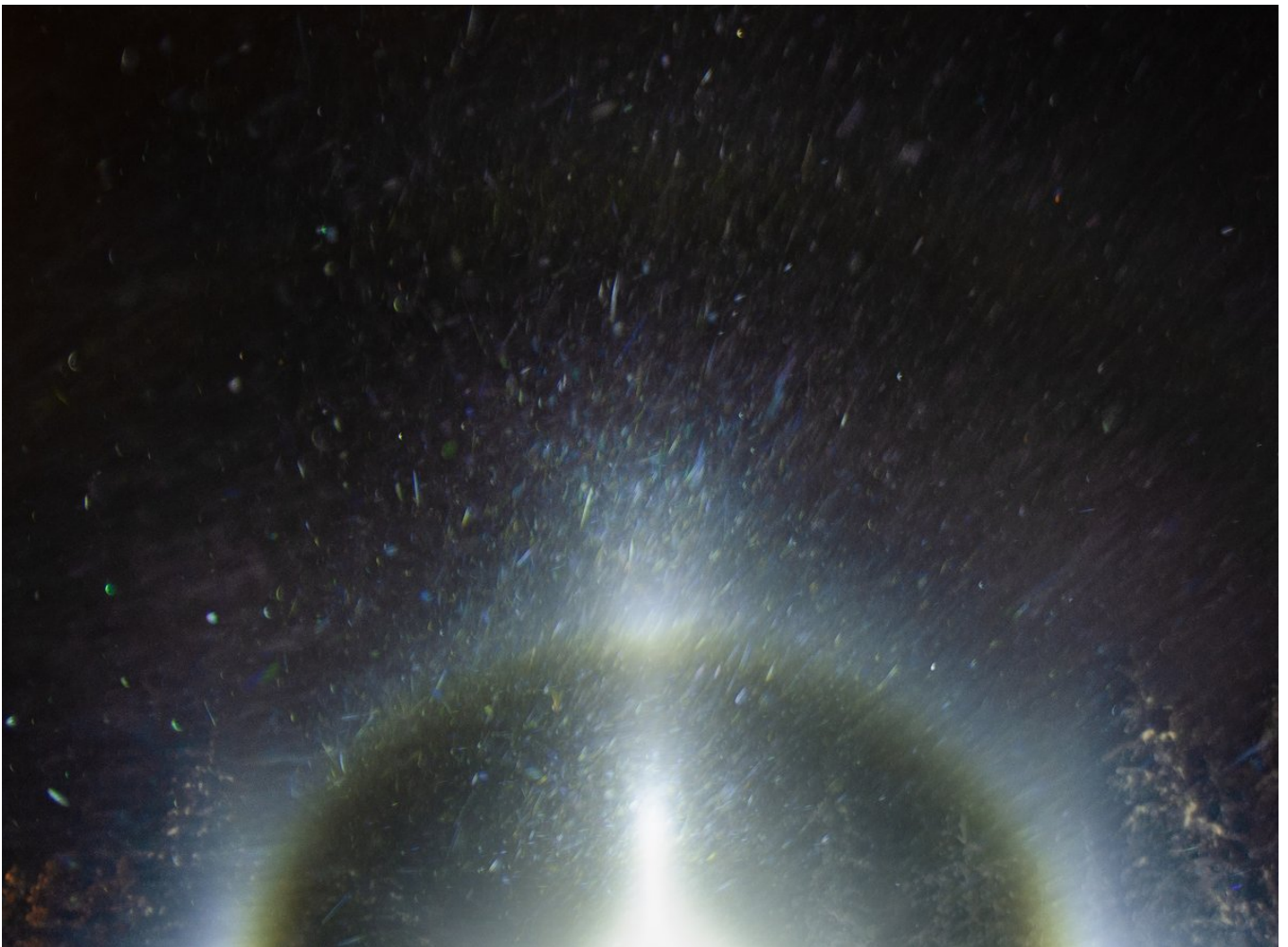


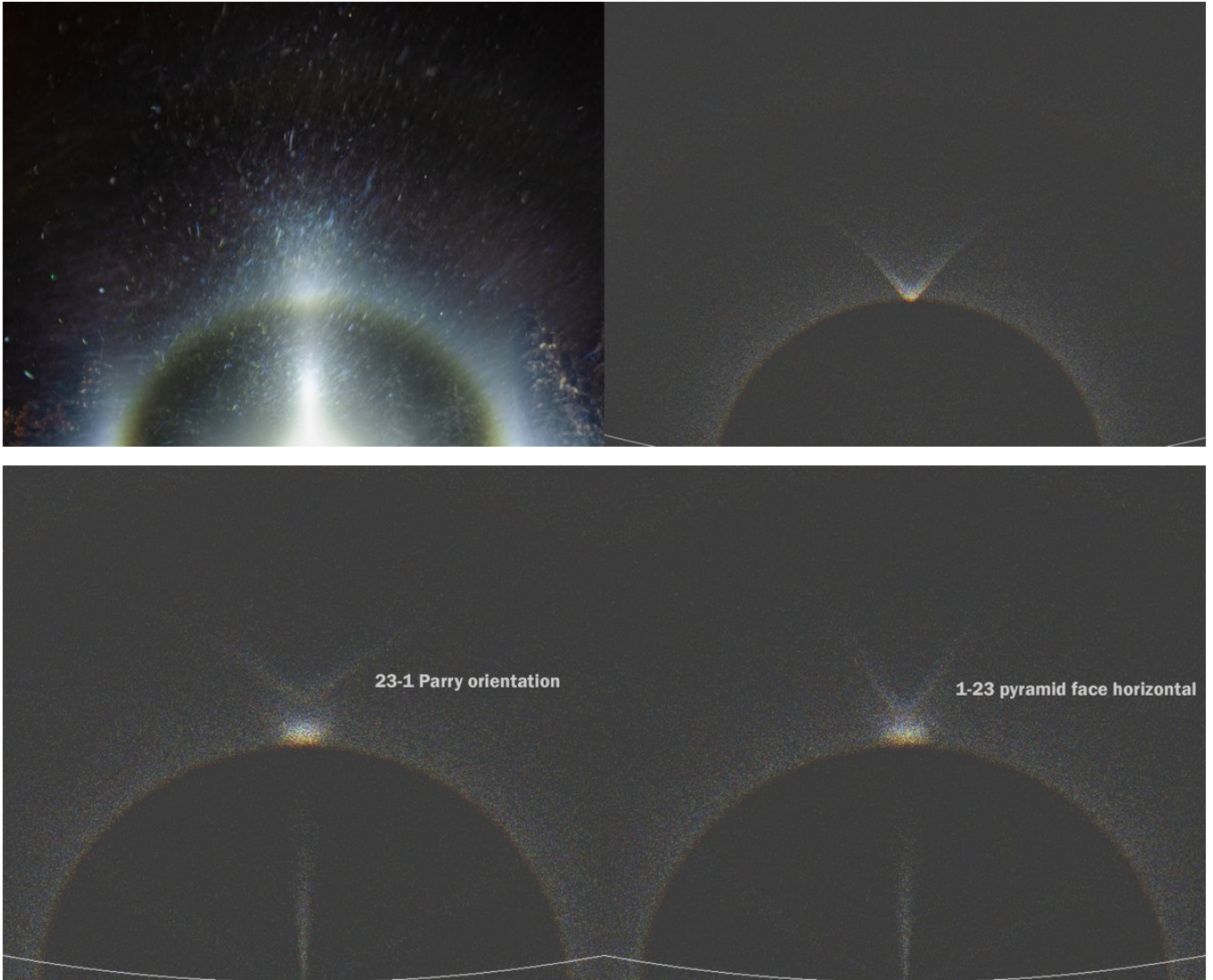
6/21 - 2026-02-21 11:29 UTC

17/18 Feb 2026, Rovaniemi. In this max stack are the frames which I estimated has the anomalous feature above the tanarc best visible (21f 9771-9799 not incl. 9781-9787). A simulation of tanarc + upper vex Parry combo is provided for comparison in the third image. Clearly, vex Parry arc is not an option, it is too close to the light. I have considered a couple of other avenues for an explanation.

- 1) Stack misalignement. Sometimes this has caused "new" halos in stacks, but it is ruled out here.
- 2) Exotic alternate Parry orientation. Give the right rotation and you pretty much get a right shaped arc in right location.
- 3) Some kind of a 23° arc. In the fourth image is shown two scenarios, Parry oriented and pyramid face horizontally oriented pyramids (the latter has been considered by Können, I don't know if he published the idea anywhere). The other arcs are filtered out, meaning scenarios are not realistic. The orientation is rather Lowitz than Parry because I used 2° degs deviation for "C-axis" and 0.5° for "Rotation" in HaloPoint.
- 4) Contact arc from exotic wedge angle. If the feature seen here is actually present also in the earlier display <https://t.co/RddZsryvsV> which has significantly different light elevation, then contact arc explanation (i.e. spinning crystals) is needed to maintain the location and shape. That's the idea at least, I have not tried to simulate.

Light elevation -1 degs for simulations.



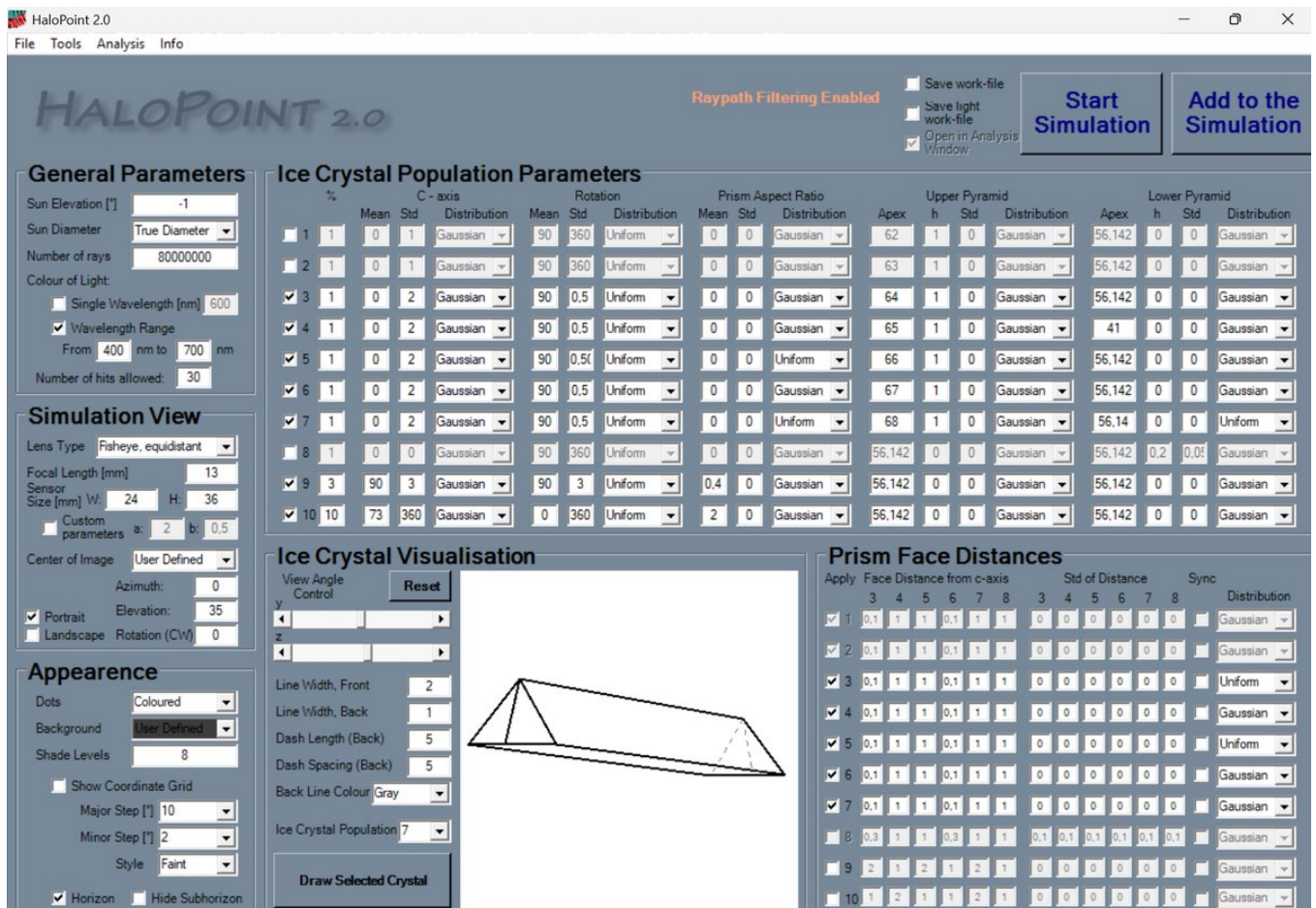
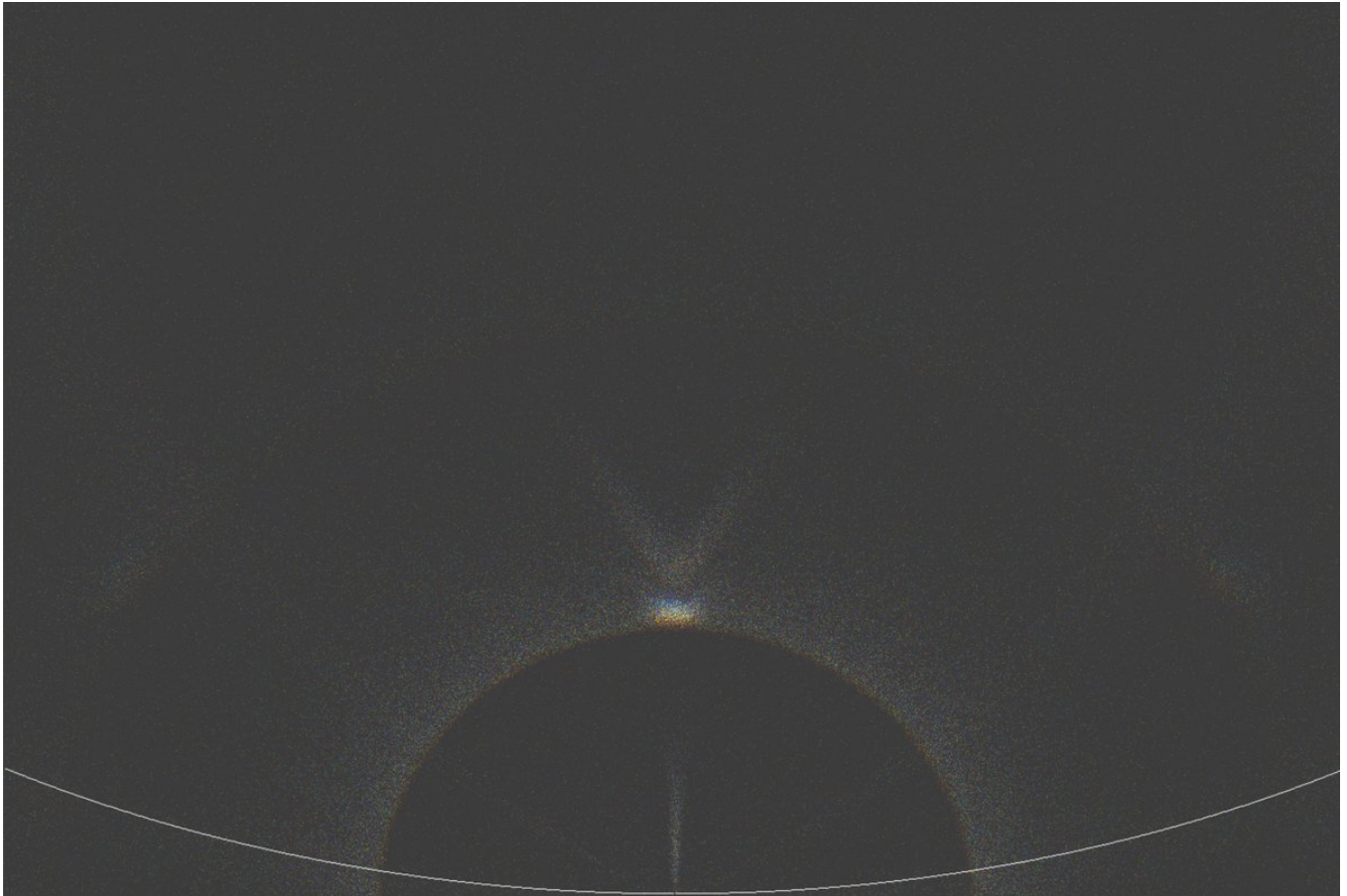


7/21 - 2026-02-21 16:18 UTC

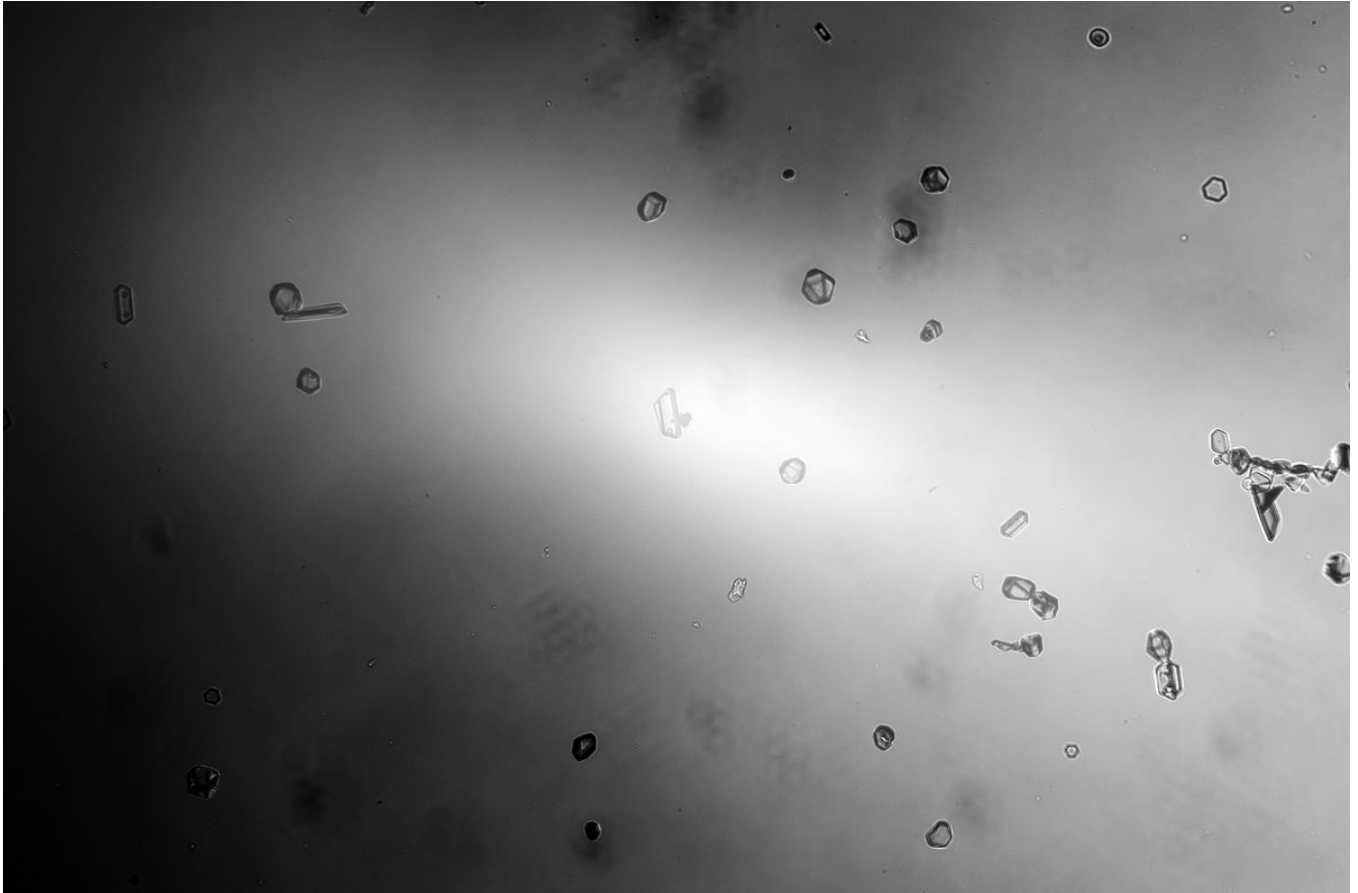
17/18 Feb 2026, Rovaniemi. The vertical extent of the feature bothers me. So here is one more idea based on crossed crystals. Crossed plates are common in snow guns displays and if they were crossed at a certain range of angles, an abnormally broad arc would ensue. Here I have imitated crossings at $64-68^\circ$.

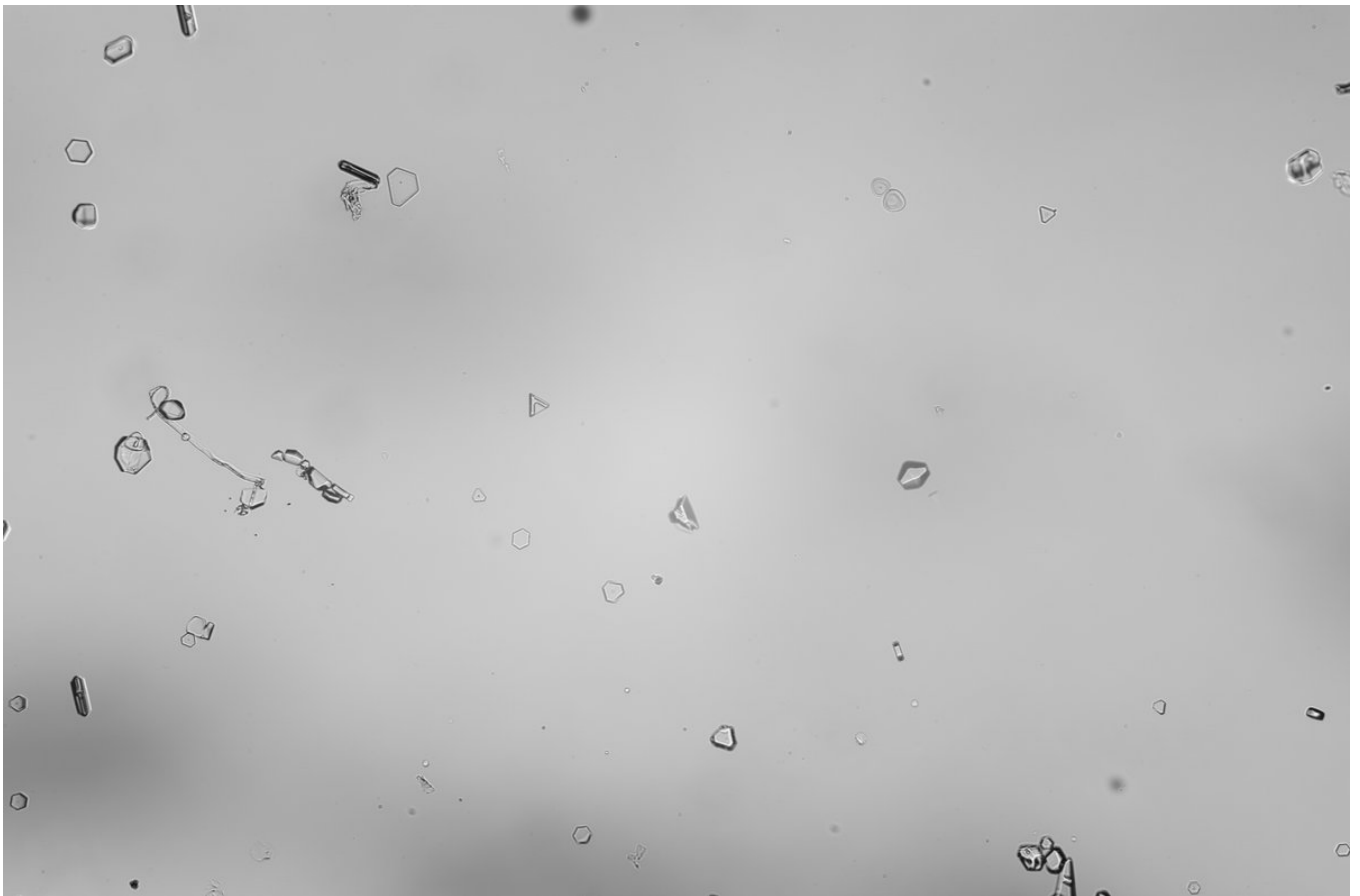
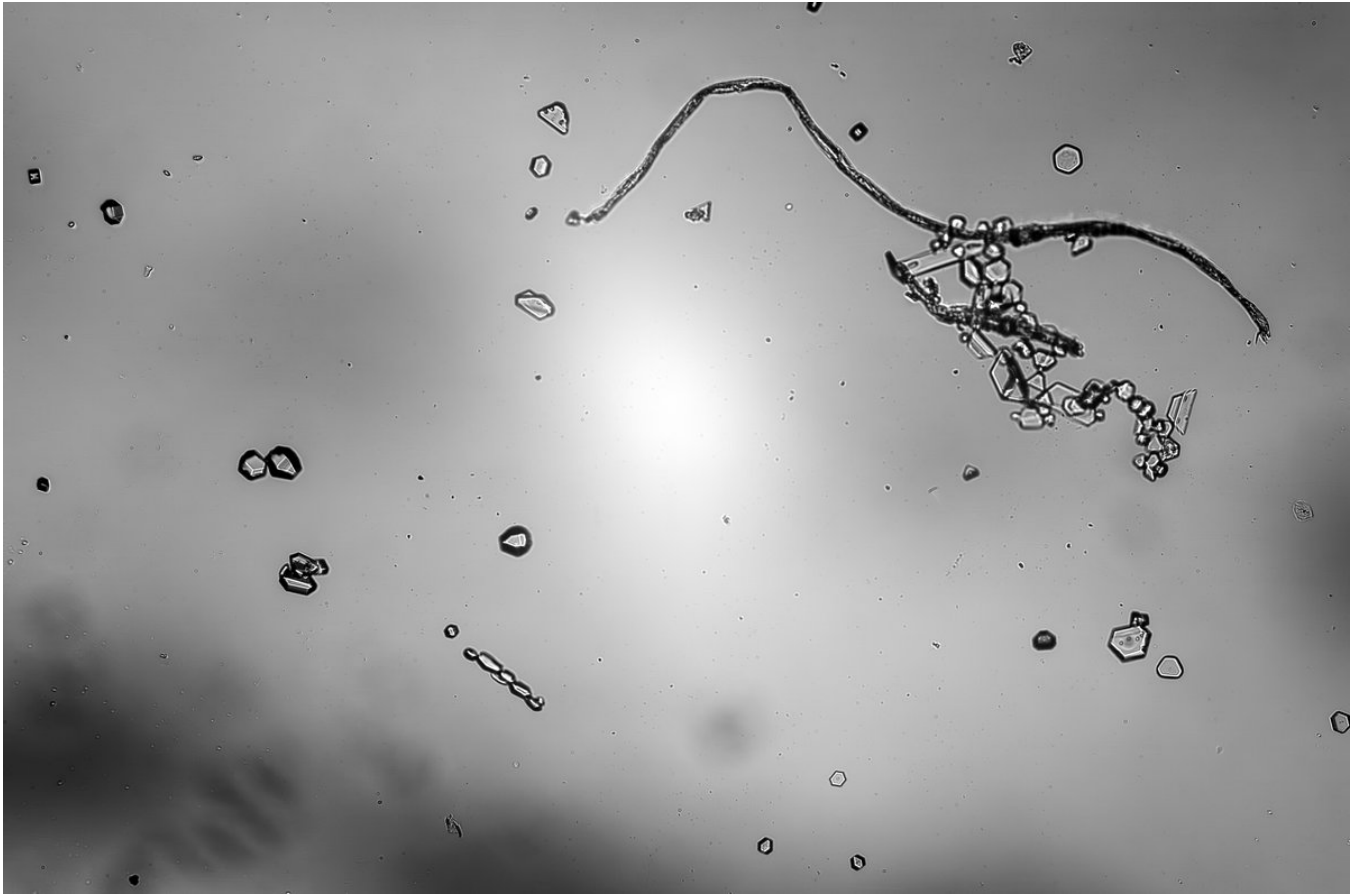
TapeMoilanen tells that they have not observed certain kind of polycrystals forming discrete angles, but a type of triple polycrystal would form discrete angles and thus could make new halos (pages 183-184).

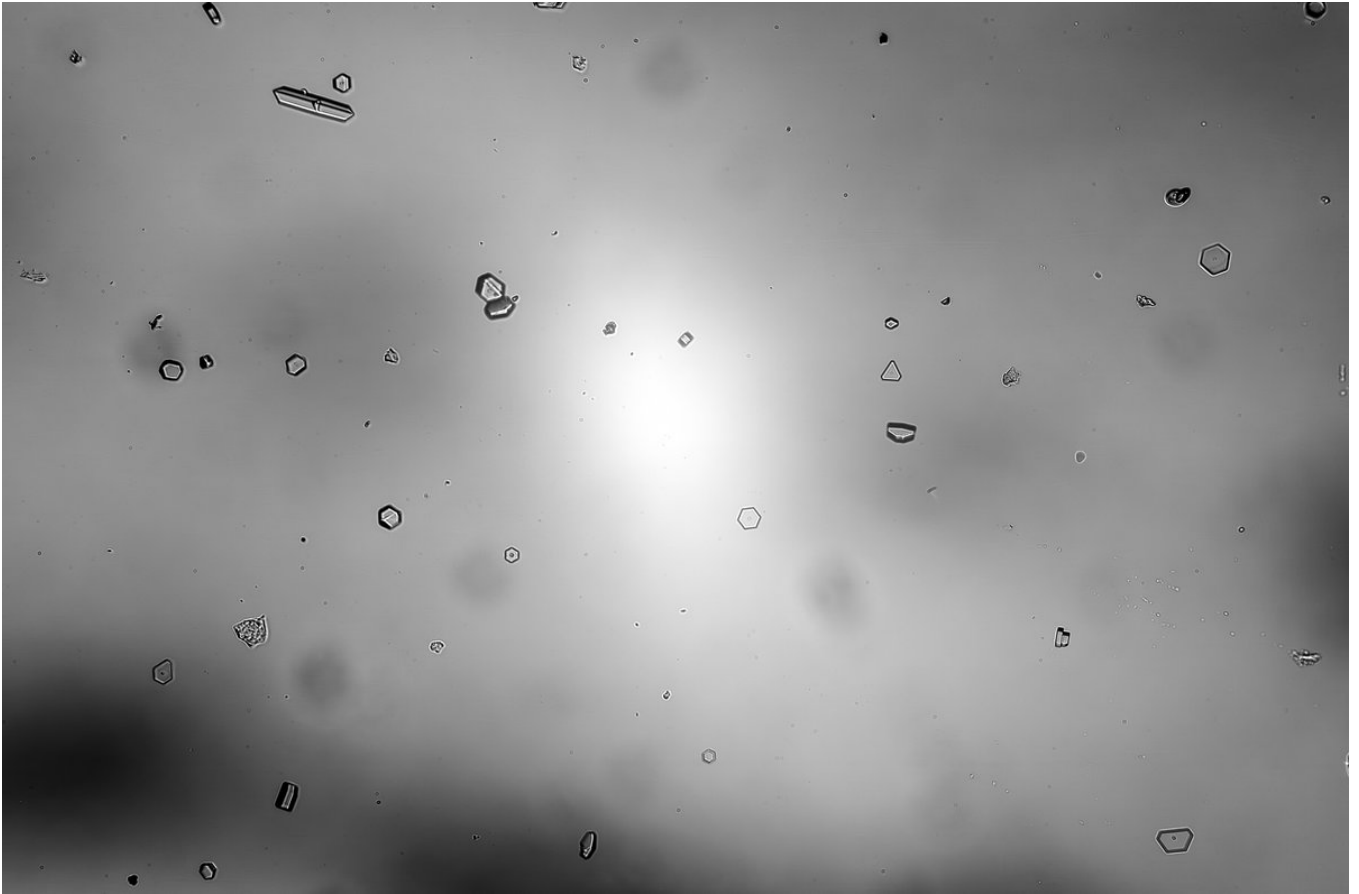
In the literature the crossing angle of $\sim 70^\circ$ is often referred, such as in "Structures and Formation Mechanisms of Snow Polycrystals" by Furukawa, figure 8 of that 1982 paper depicting such crossed plate. Weinheimer and Knight brought up crystals joined at this angle in 1987 paper "Scheiner's halo: Cubic ice or polycrystalline hexagonal ice?". Interestingly, in figure 1 of that paper there is a table by Lee of measured crossings of actual crystals, showing that there is a range of angles. The original by Lee, "On the Crystallographic Orientation of Spatial Branches in Natural Polycrystalline Snow Crystals", has a couple of nice photos of crossed plates.



17/18 Feb 2026, Rovaniemi. I put the dish cooking at 2216, which is pretty much when the 9° halo appears in photos. Photographed crystals after the swarm disappeared. Photos are not sharp, crystals rounded and pyramids in particular have dark contrast ring surrounding them, but they are good enough to identify the crystal types. Moilanen arc was present and so are the elongated crossed crystals that we think are responsible. And the strong 18, 20, 35° halos are in line with the long pyramid faces.



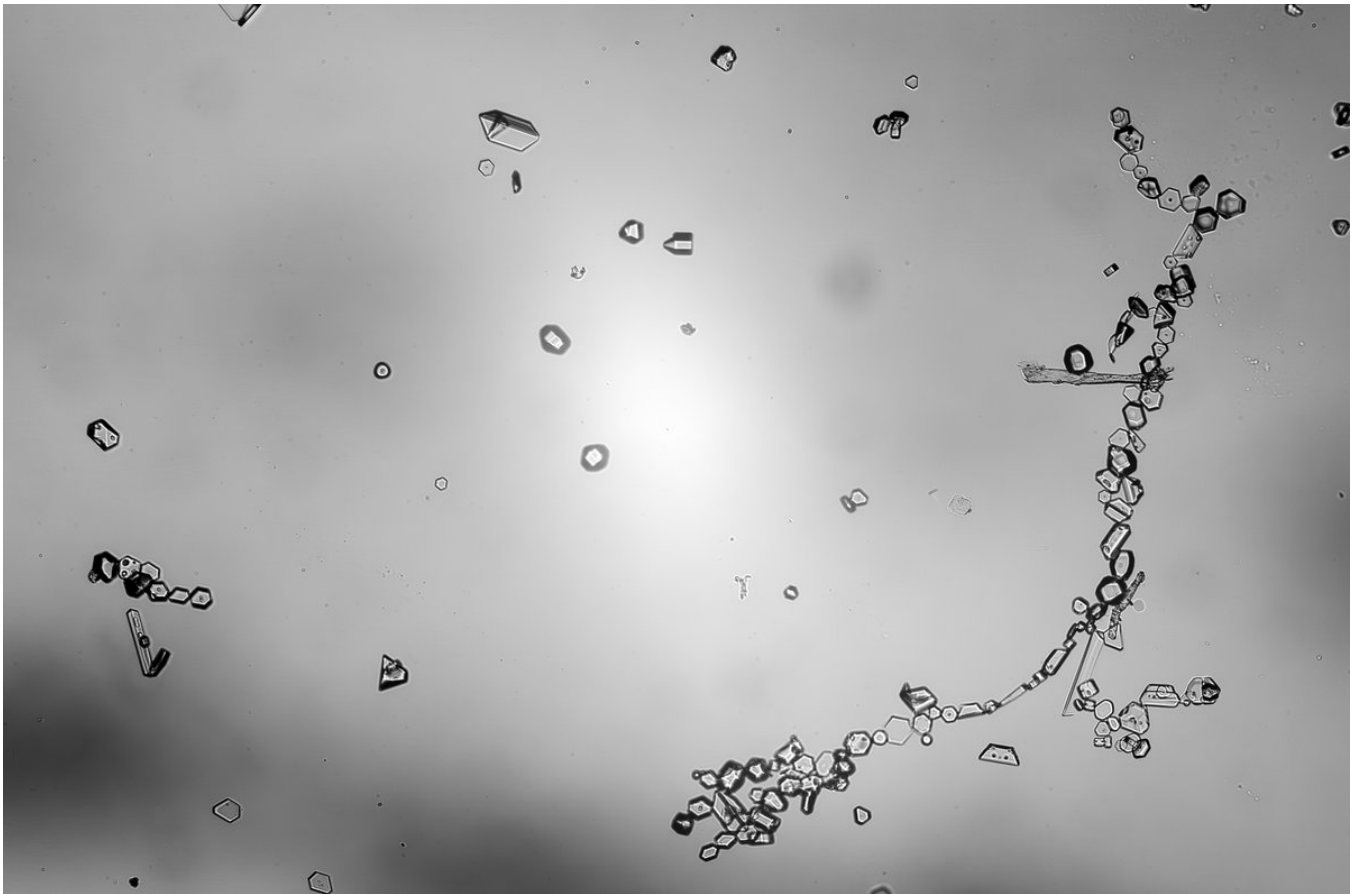




9/21 - 2026-02-22 09:37 UTC

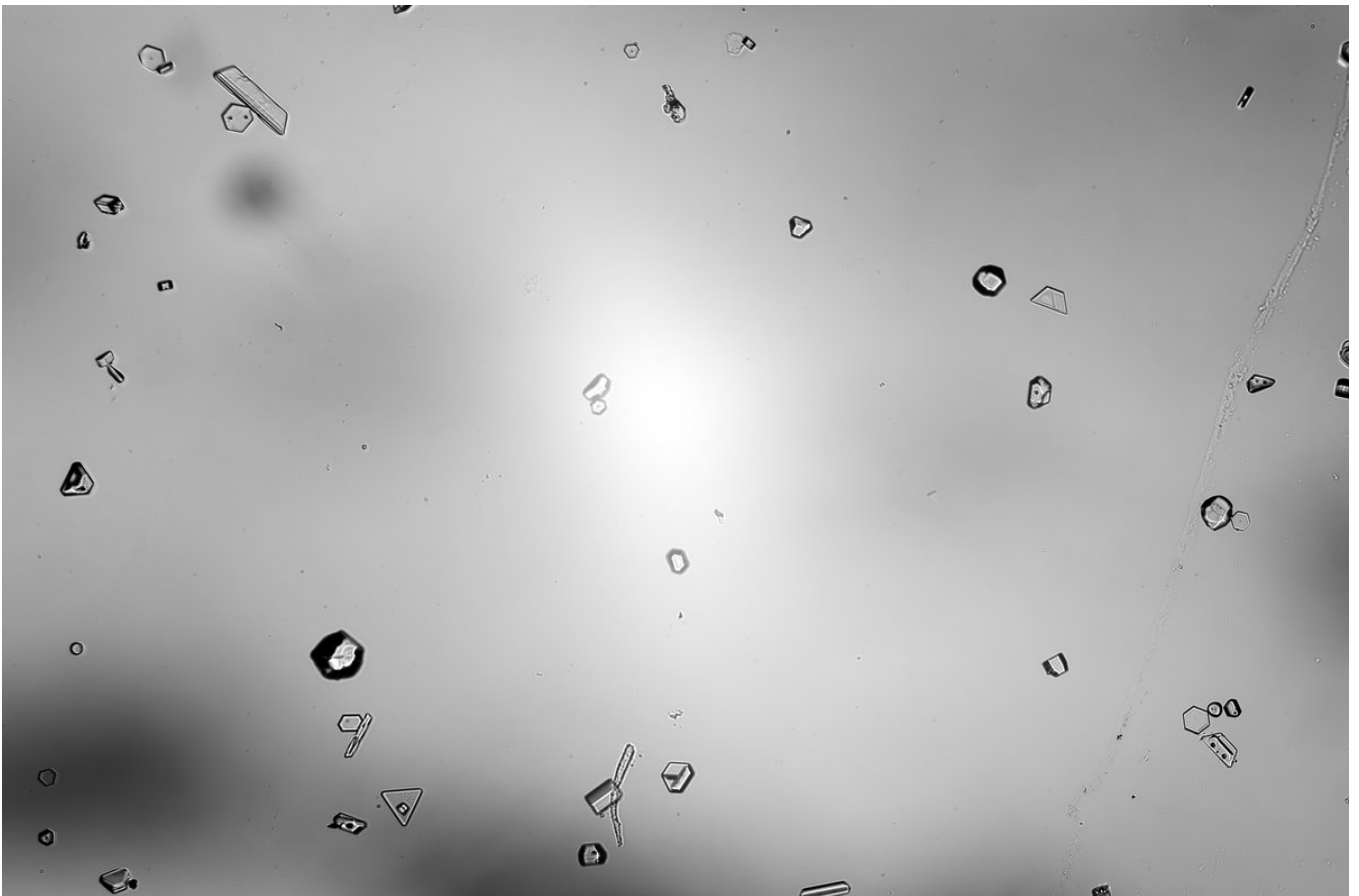
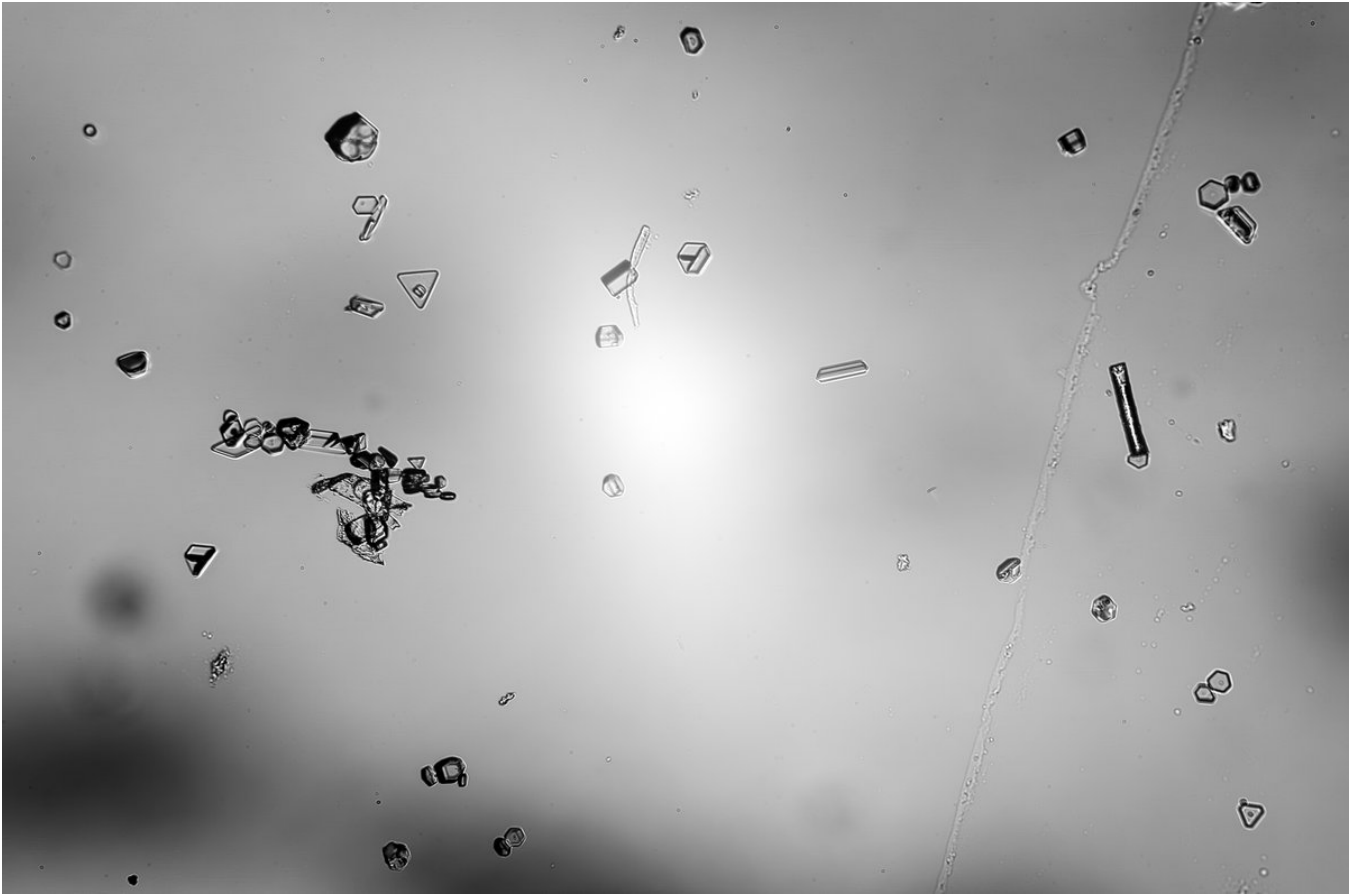
17/18 Feb 2026, Rovaniemi. The commonly seen triple crossed plate crystal is nicely displayed at the top of the third image. TapeMoilanen writes on page 184 about such type: "...the line of intersection of any to plates is parallel to an a-axis of each, and this implies that the orientations of the plates with respect to one another are fixed. Crystals like these would make new halos if they were large enough and occurred in sufficient numbers". This is in reference to an ESEM photo on the same page of triple crossed pyramid, so I don't know if non-pyramid type will also contain exotic angles. TapeMoilanen has a few more ESEMs of triple crossed crystals, on page 111 and 164.

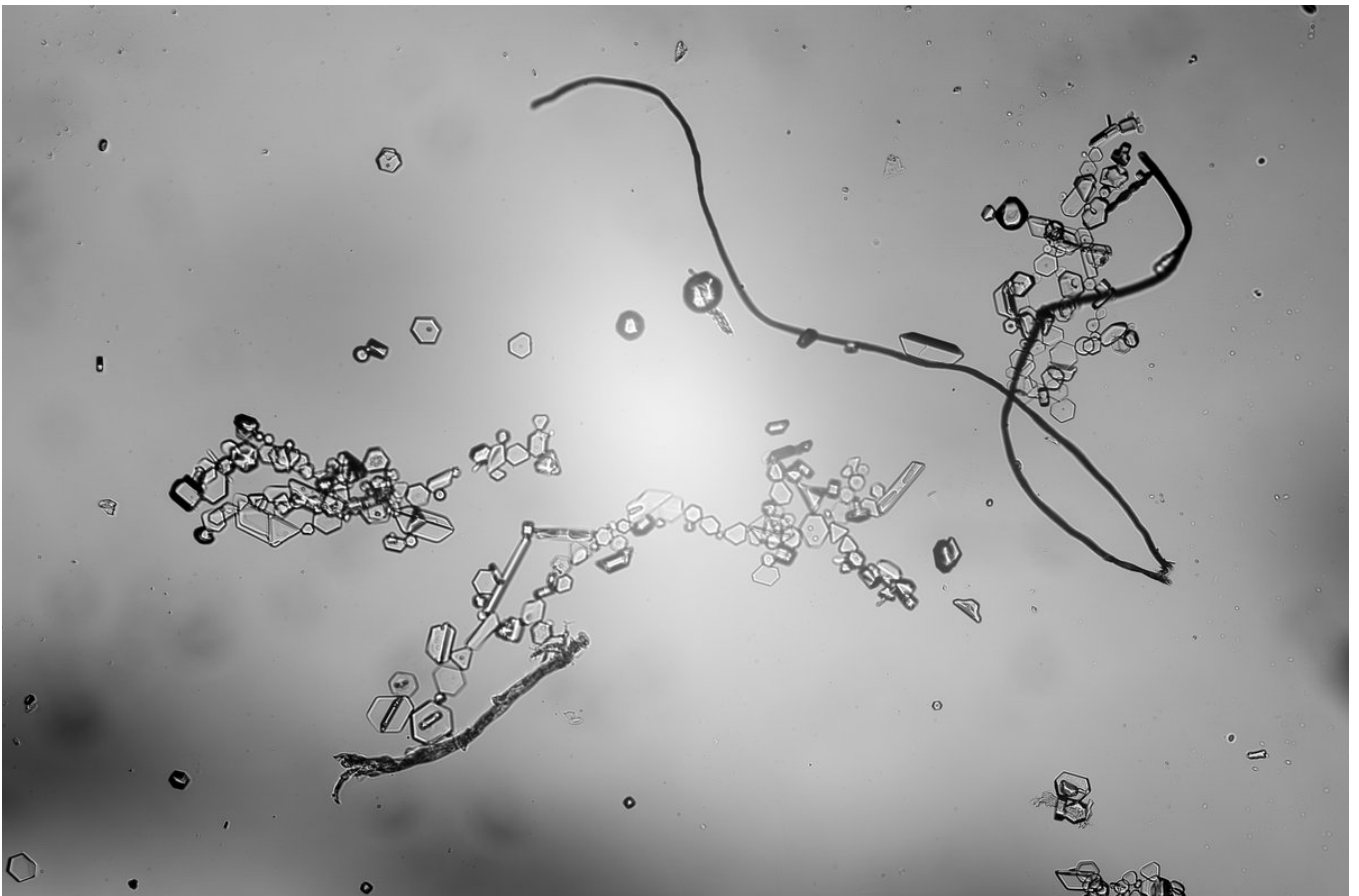
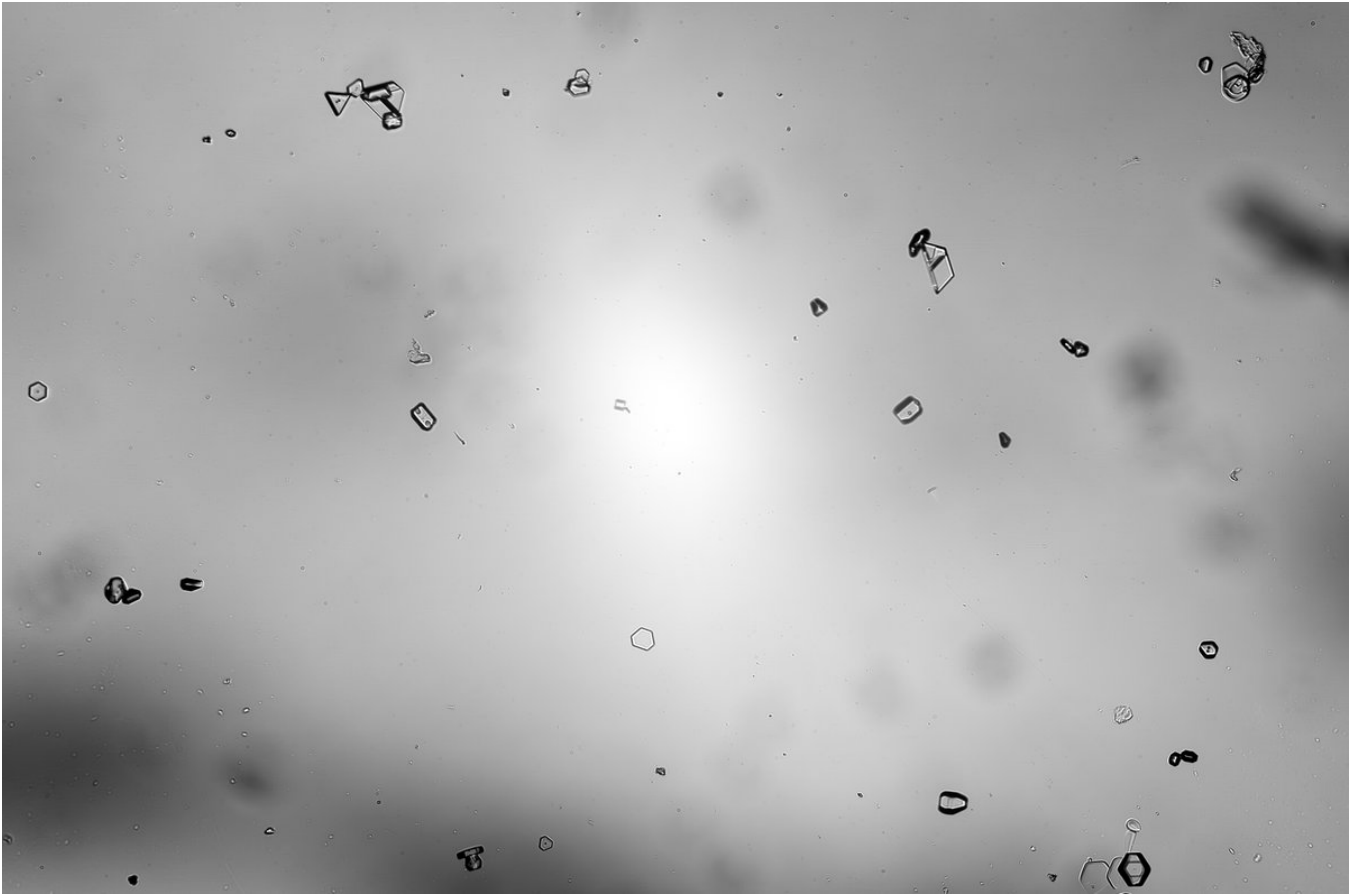




10/21 - 2026-02-22 09:38 UTC

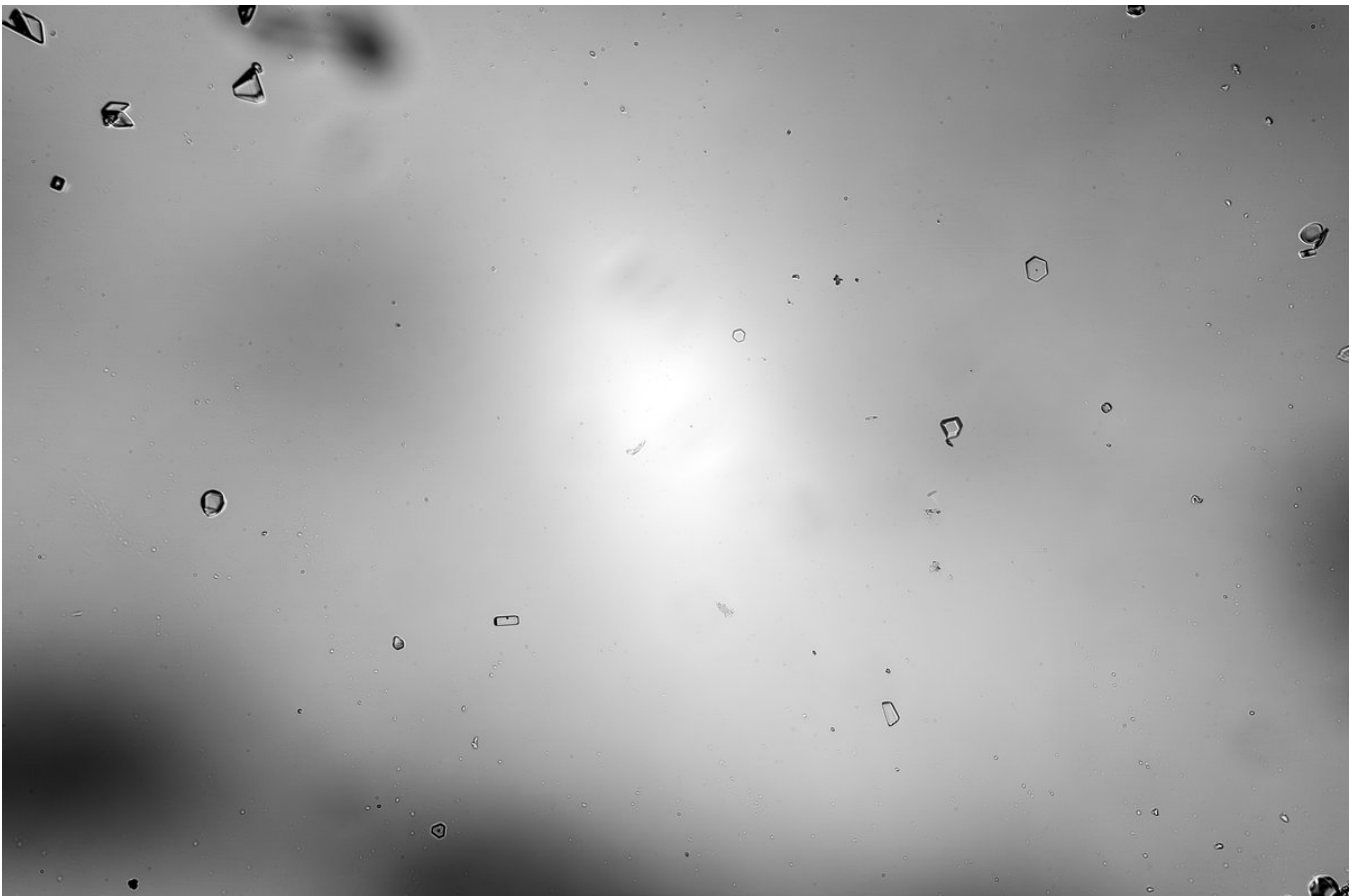
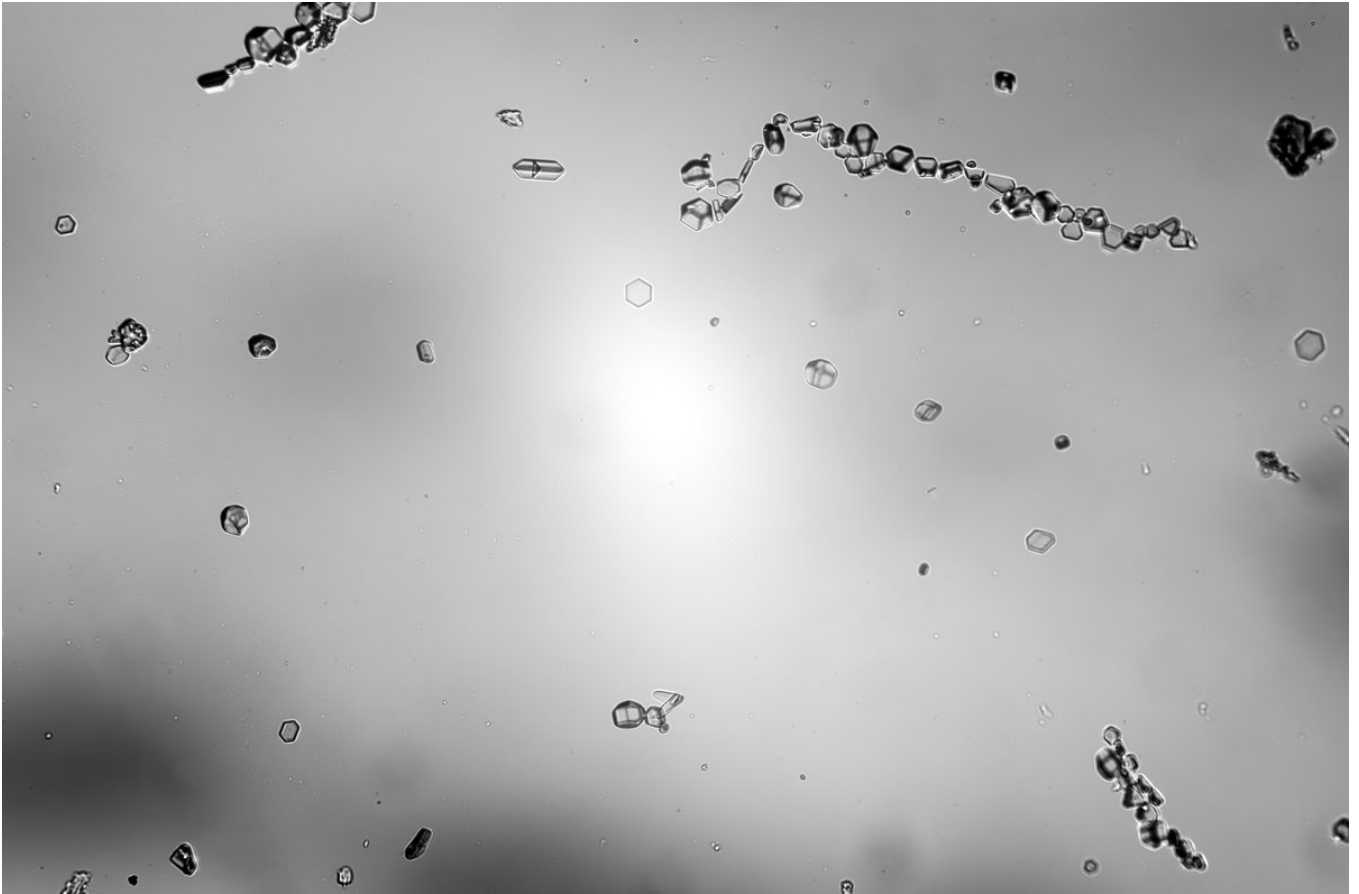
17/18 Feb 2026, Rovaniemi. More crystals.





11/21 - 2026-02-22 09:44 UTC

17/18 Feb 2026, Rovaniemi. More crystals. The big crystal is from natural precipitation.





12/21 - 2026-02-22 10:32 UTC

17/18 Feb 2026, Rovaniemi. Of the three swarms the second disappeared quickly and got no photos but the third one lasted long enough also for a crystal sample. The first frame in this 17 x 30s stack is from the time I put the dish out. I have more photos, but the swarm started withering away.

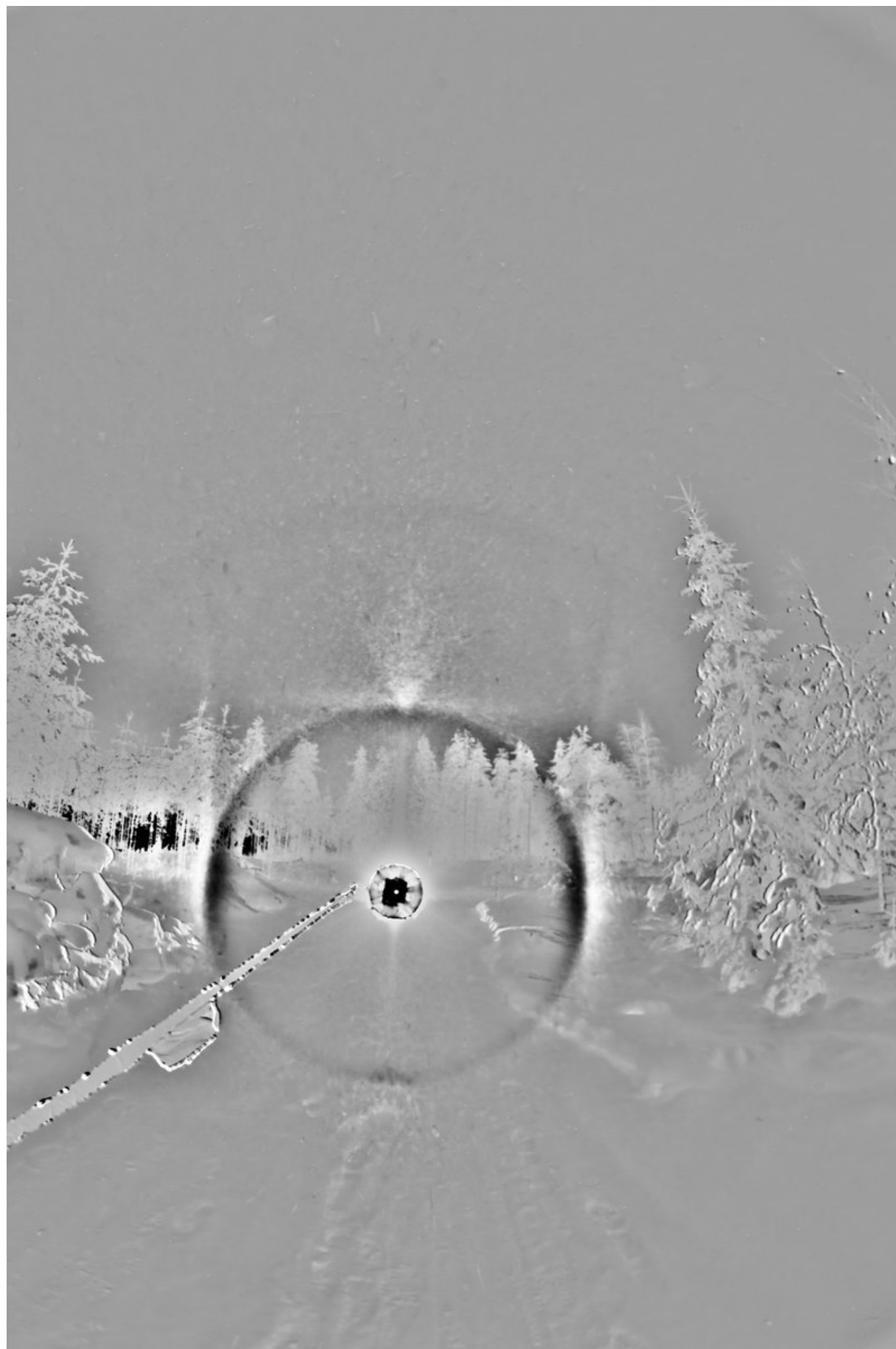
The b-g would seem to reveal lower vex Parry. It is interesting to contrast this with the display two nights later <https://t.co/mpUm5TjegU> where nothing is seen below the 22° halo despite less light polluted location (the lamp elevation is such that probably lower shallowcave Parry would be expected instead of lower vex Parry). If possible, it is good to include always both above and below 22° halo areas to have extra information on what's going on with Parry / tanarc.

No Moilanen arc seem to be present, which is interesting with regard to the crystals that are shown in the next posts.

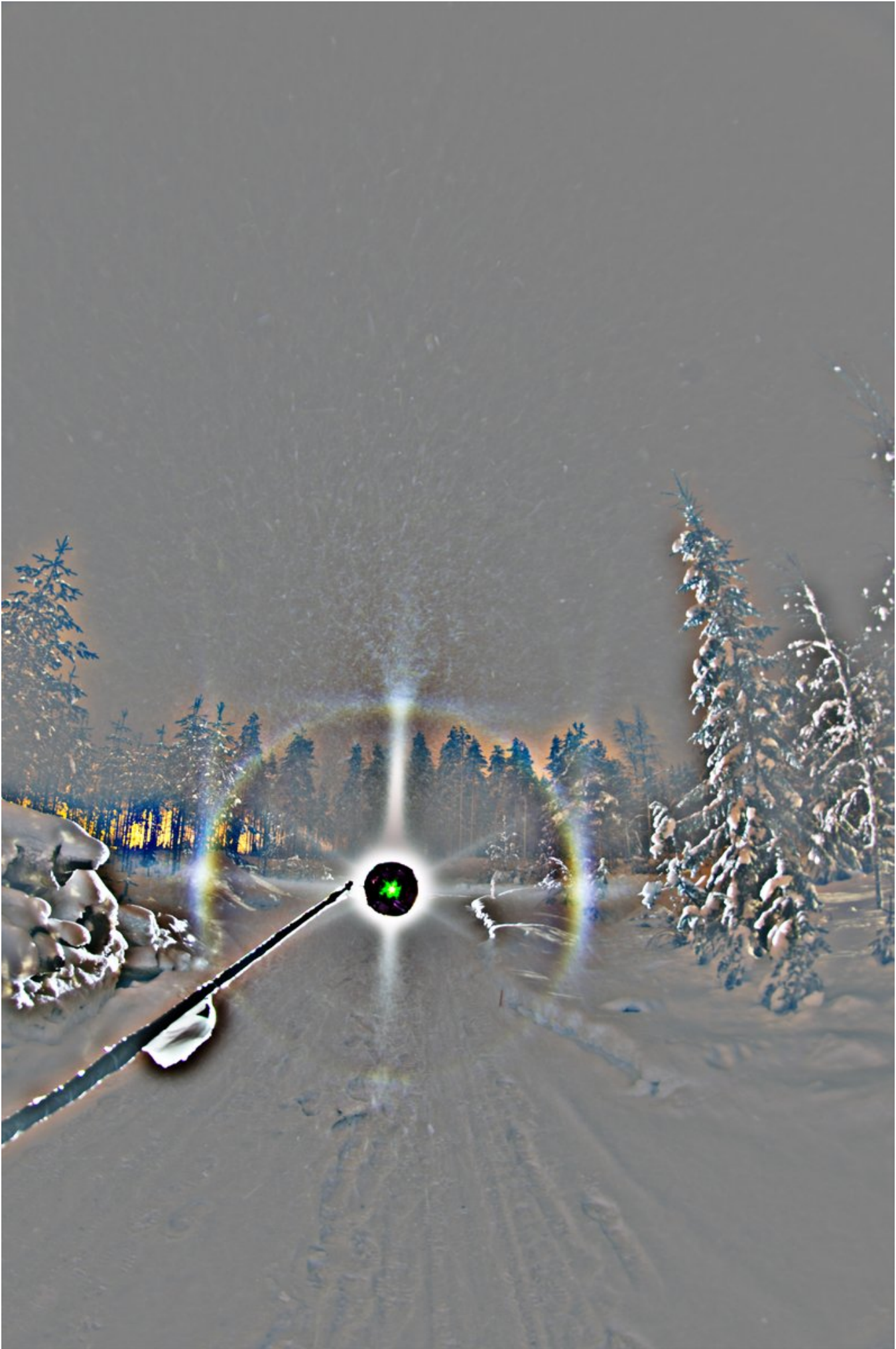
The Hastings/Wegener is short but I get vibes of the pseudo type.

Unlike in the first stage, where the lamp was on the car roof, here it was on the ground and thus further from the horizon (camera is also closer to the lamp here). The lower vex Parry allows for an estimate, maybe 5 degrees or so.



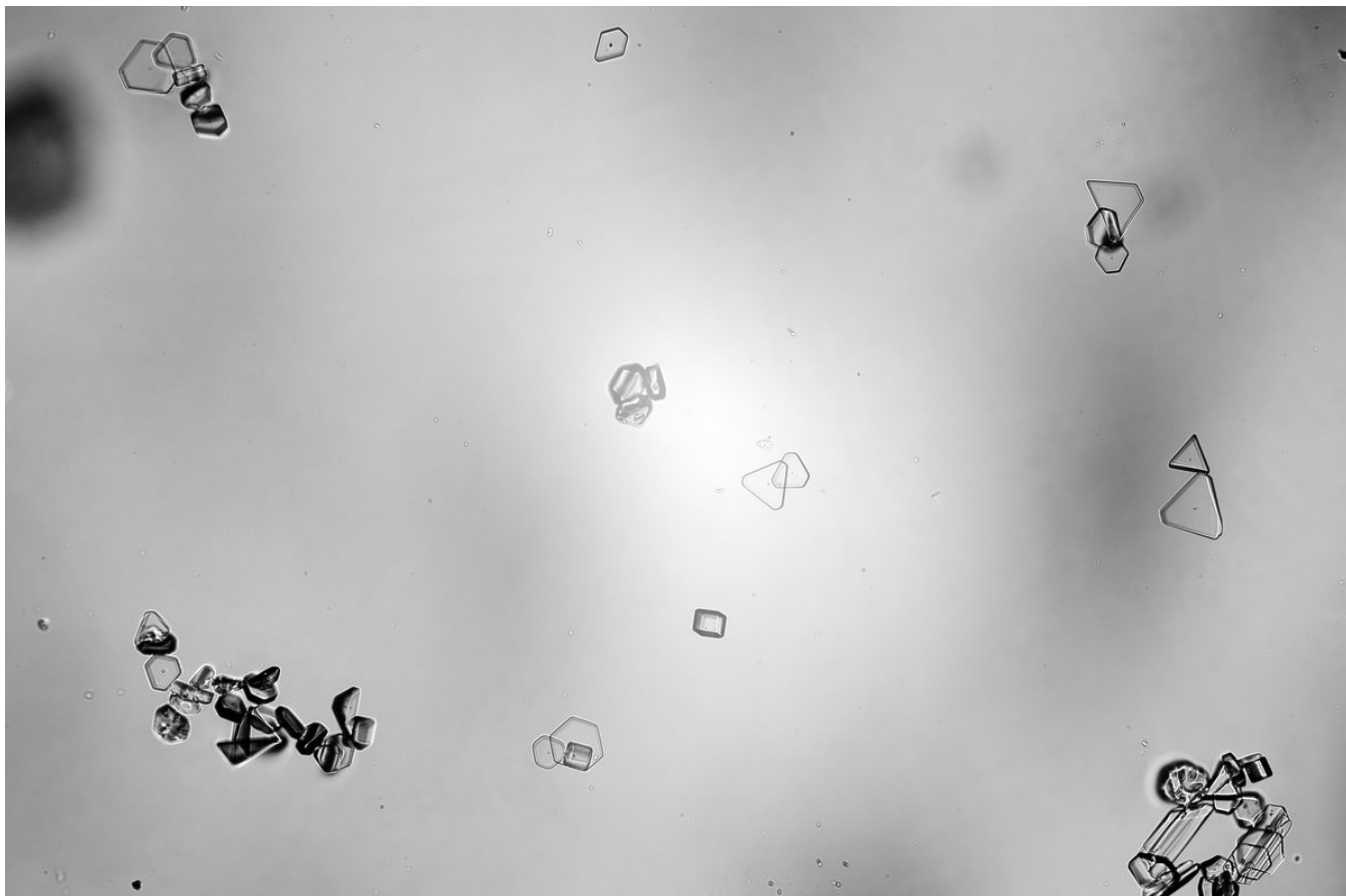


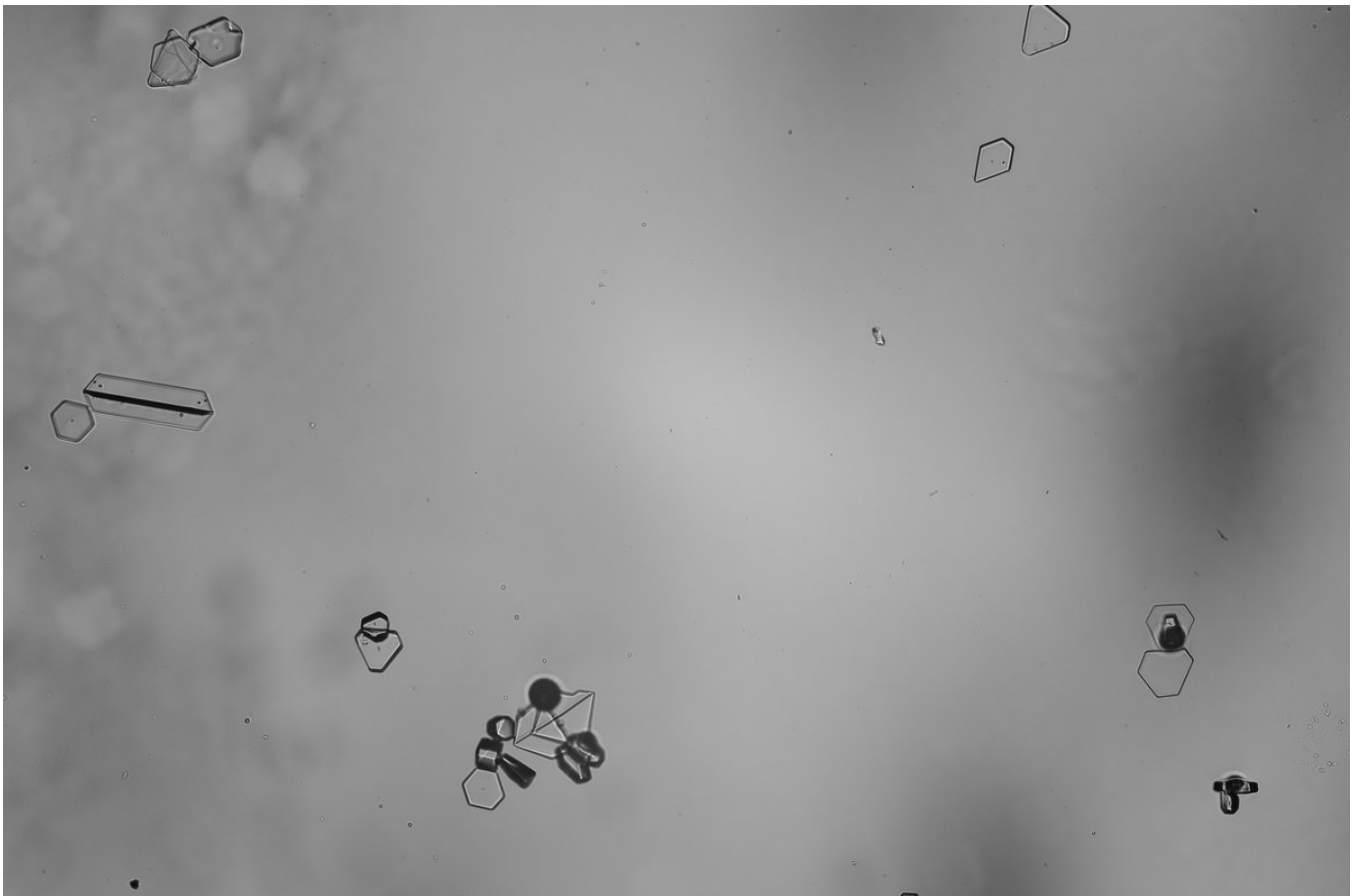
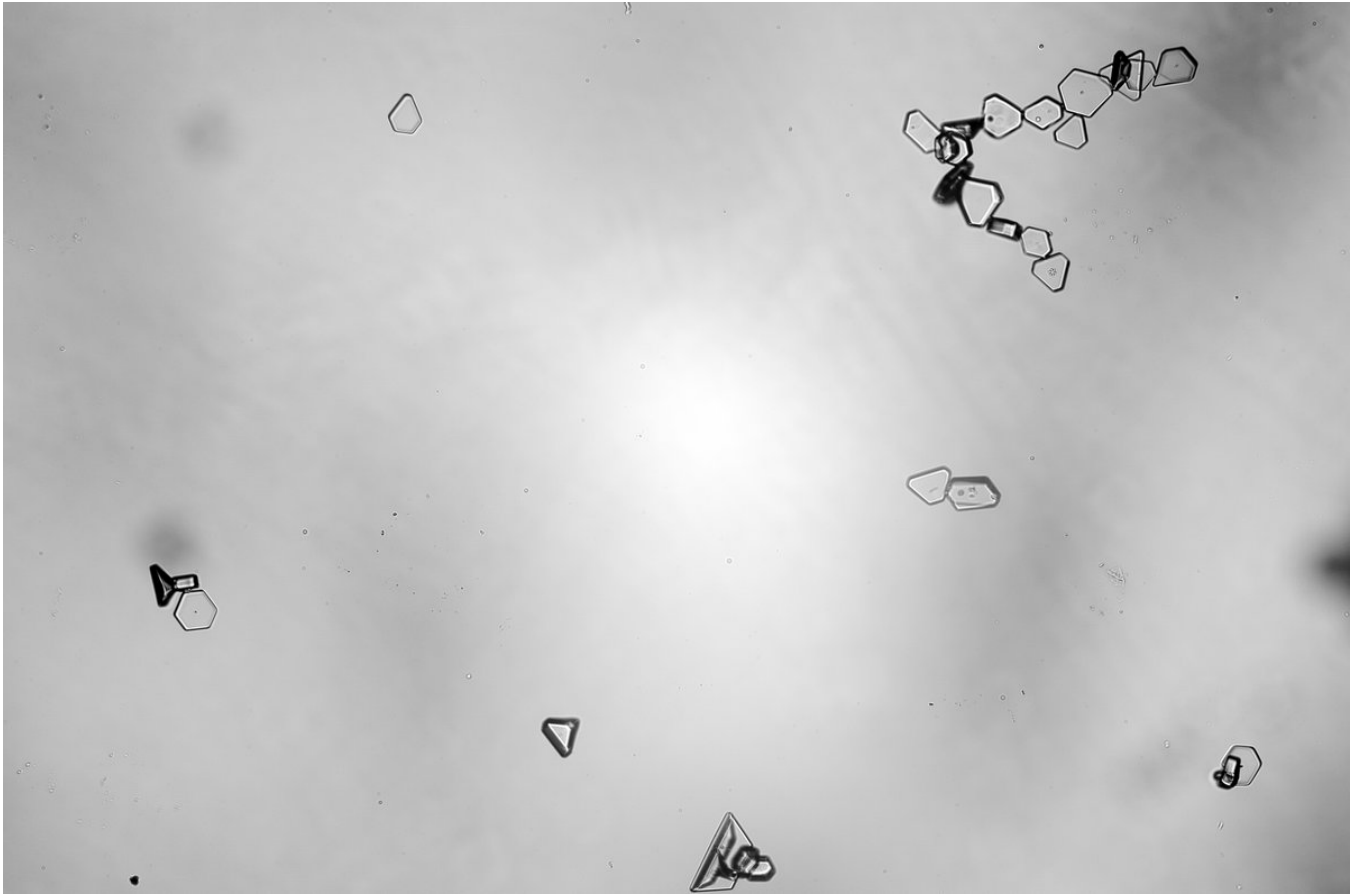


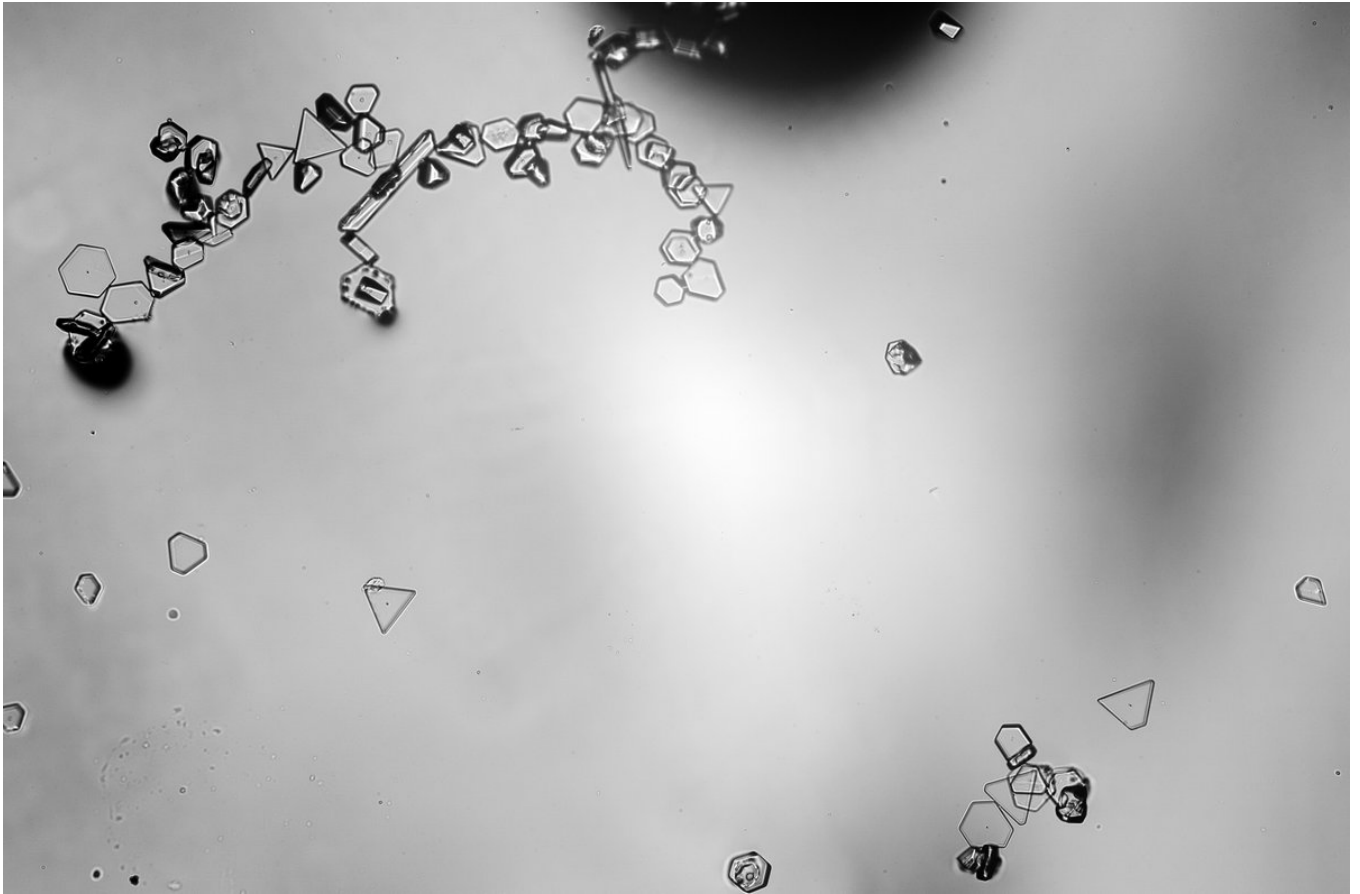


13/21 - 2026-02-22 11:16 UTC

17/18 Feb 2026, Rovaniemi. Crystals of the third swarm. You need an elongated wedge to make long wings of Moilanen arc and the elongated crossed plates fit the bill. Looking just at the sample I would say "Moilanen display" without hesitation. Yet no Moilanen arc is visible in the above post's stack.

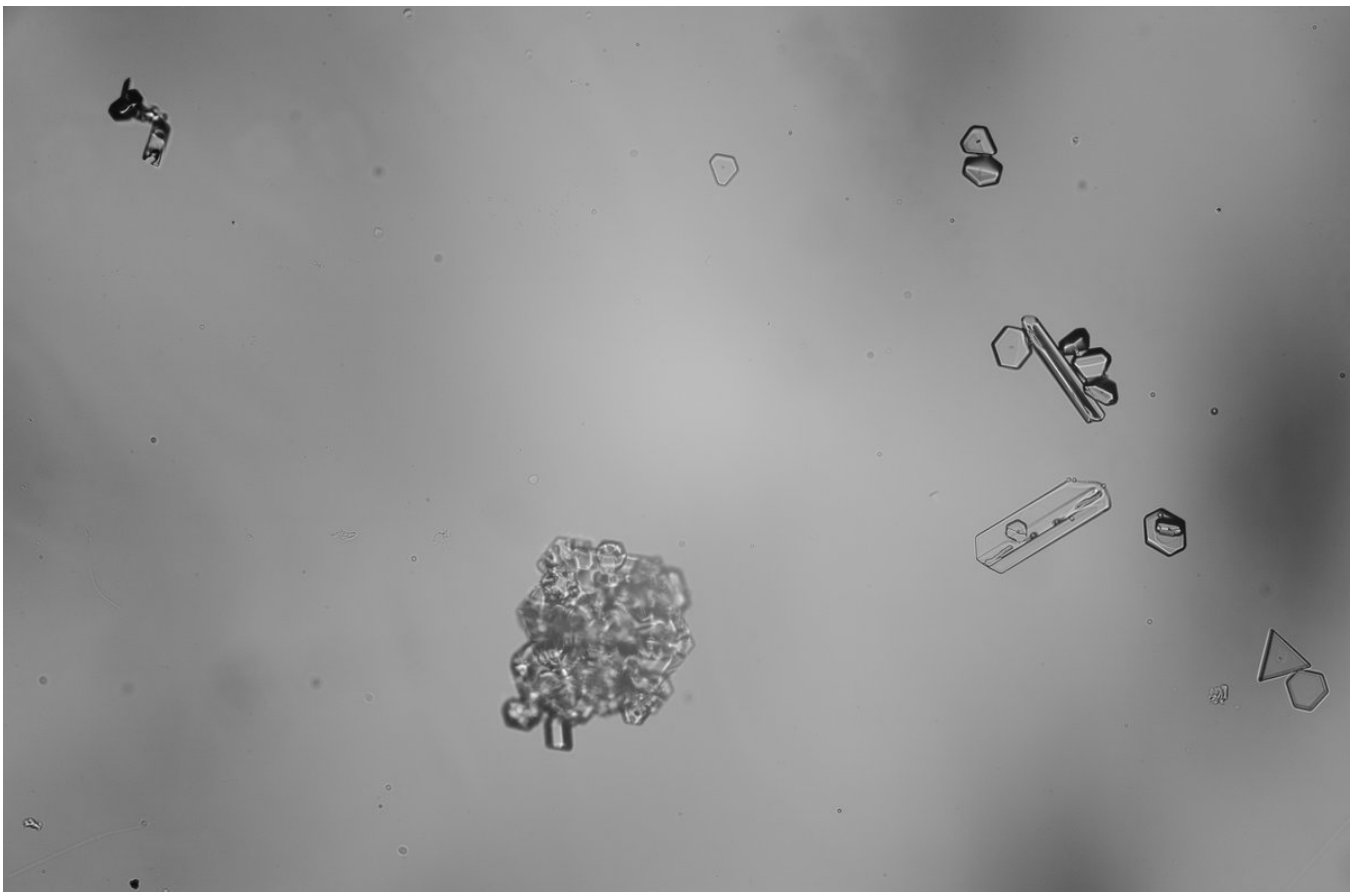


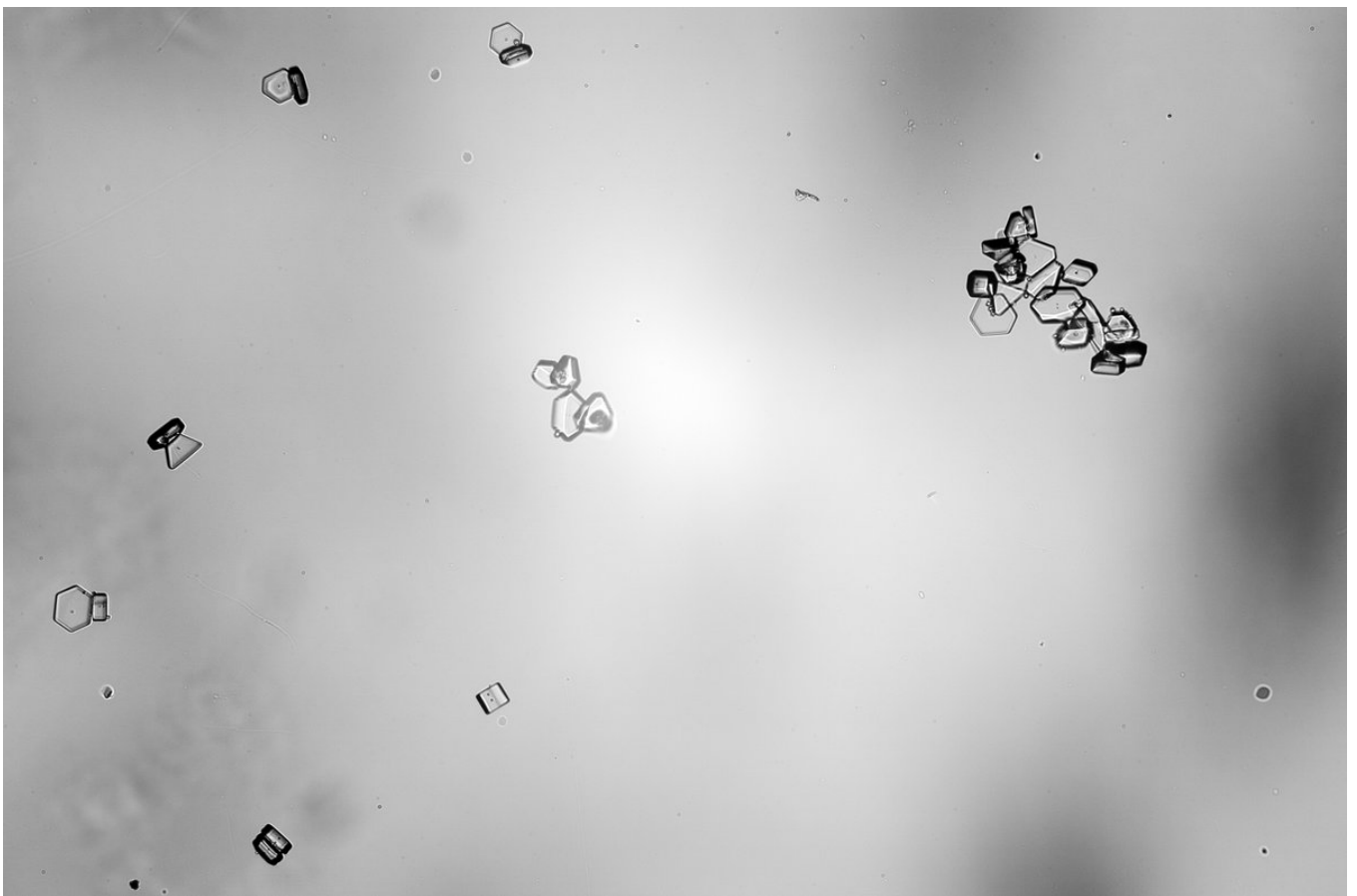
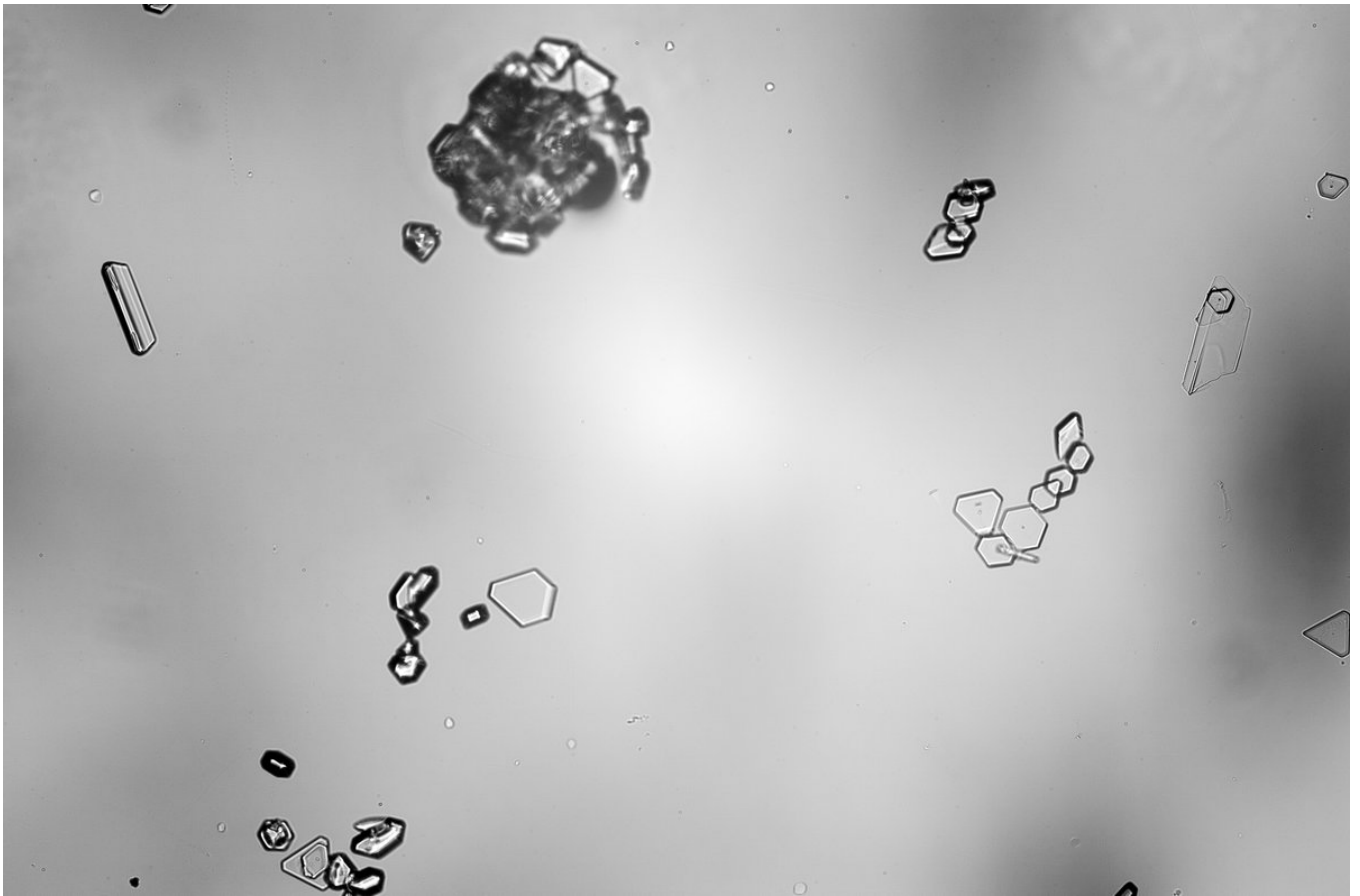


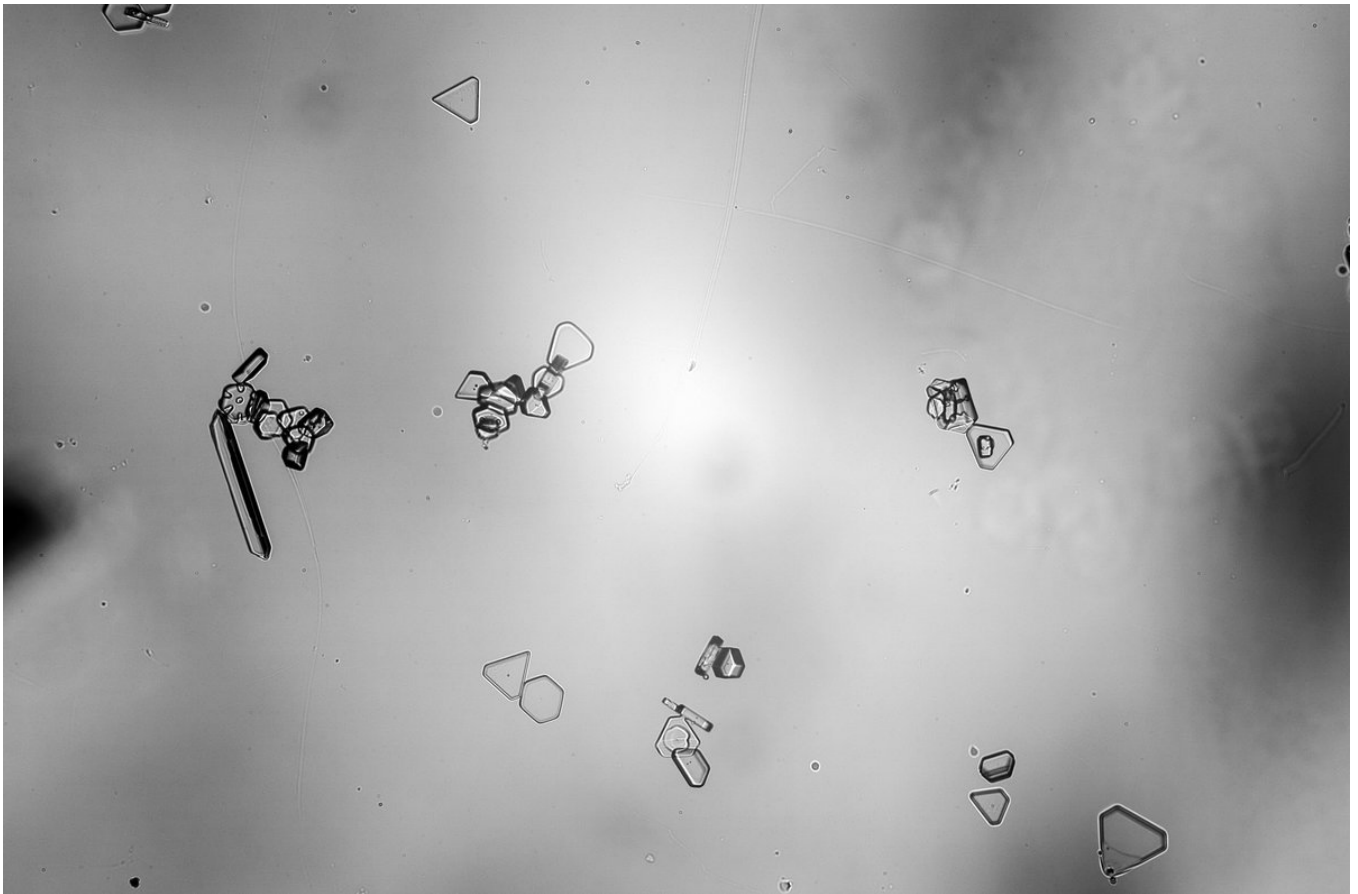


14/21 - 2026-02-22 11:16 UTC

17/18 Feb 2026, Rovaniemi. Crystals of the third swarm.

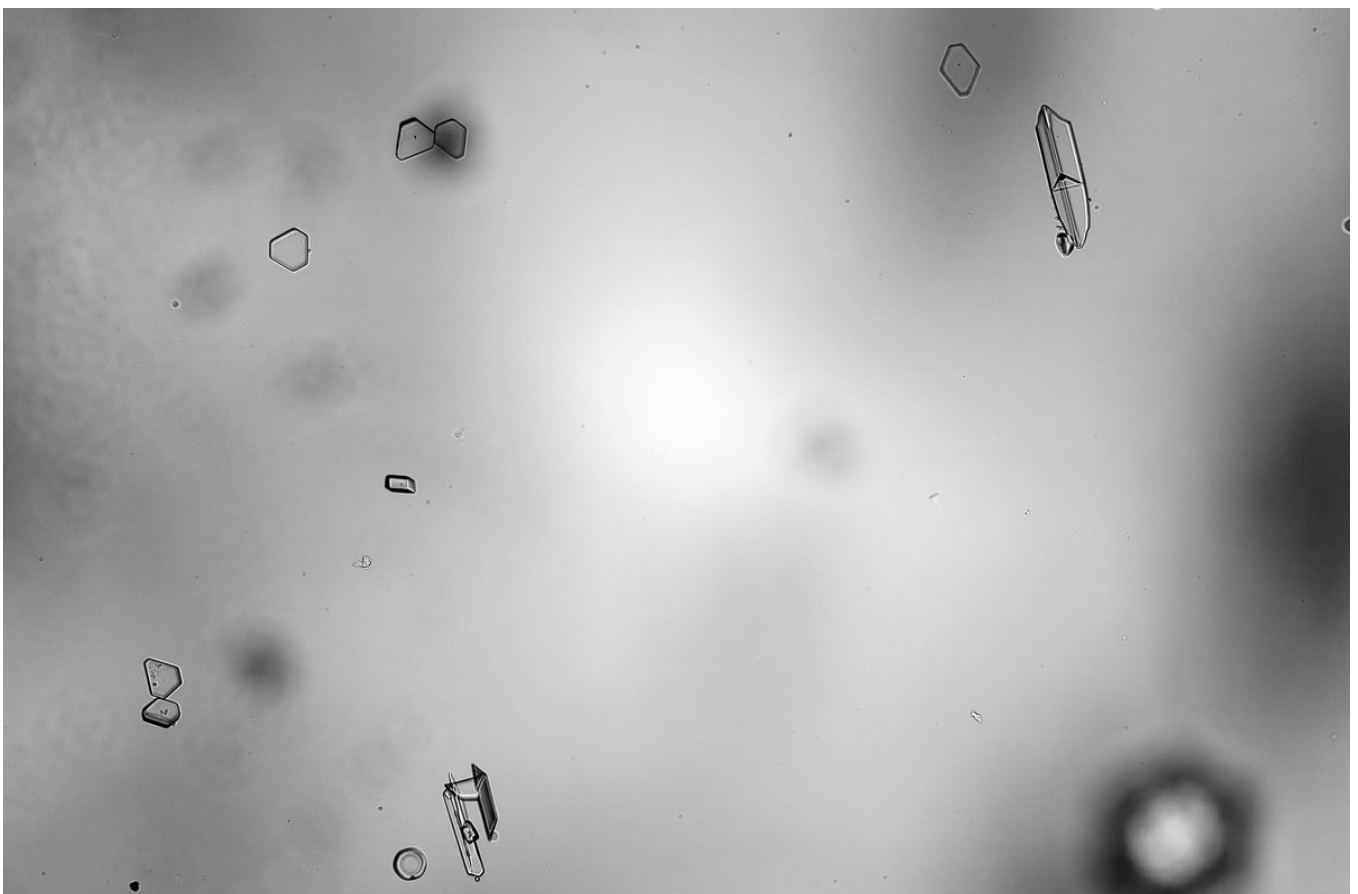


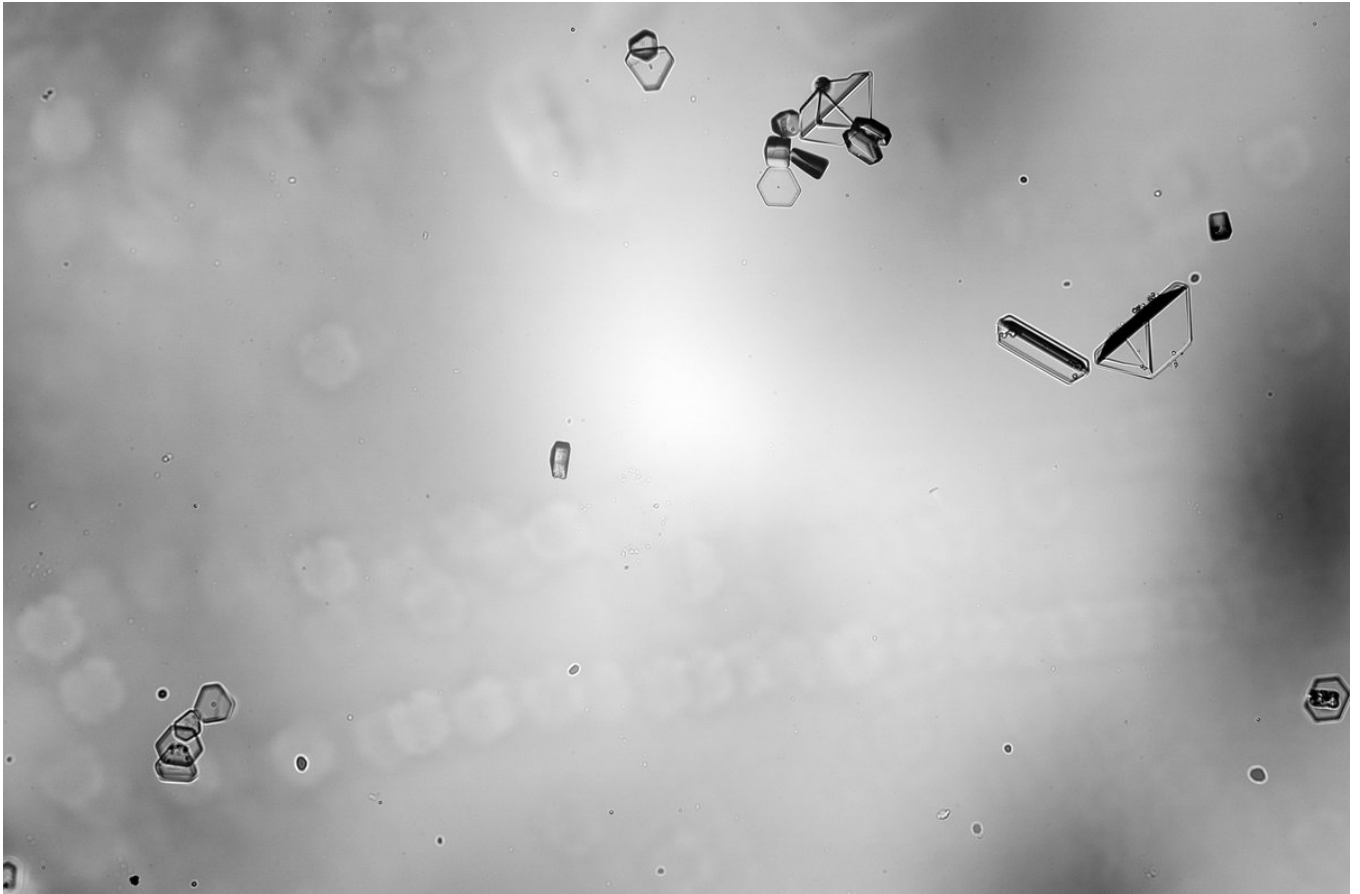




15/21 - 2026-02-22 11:17 UTC

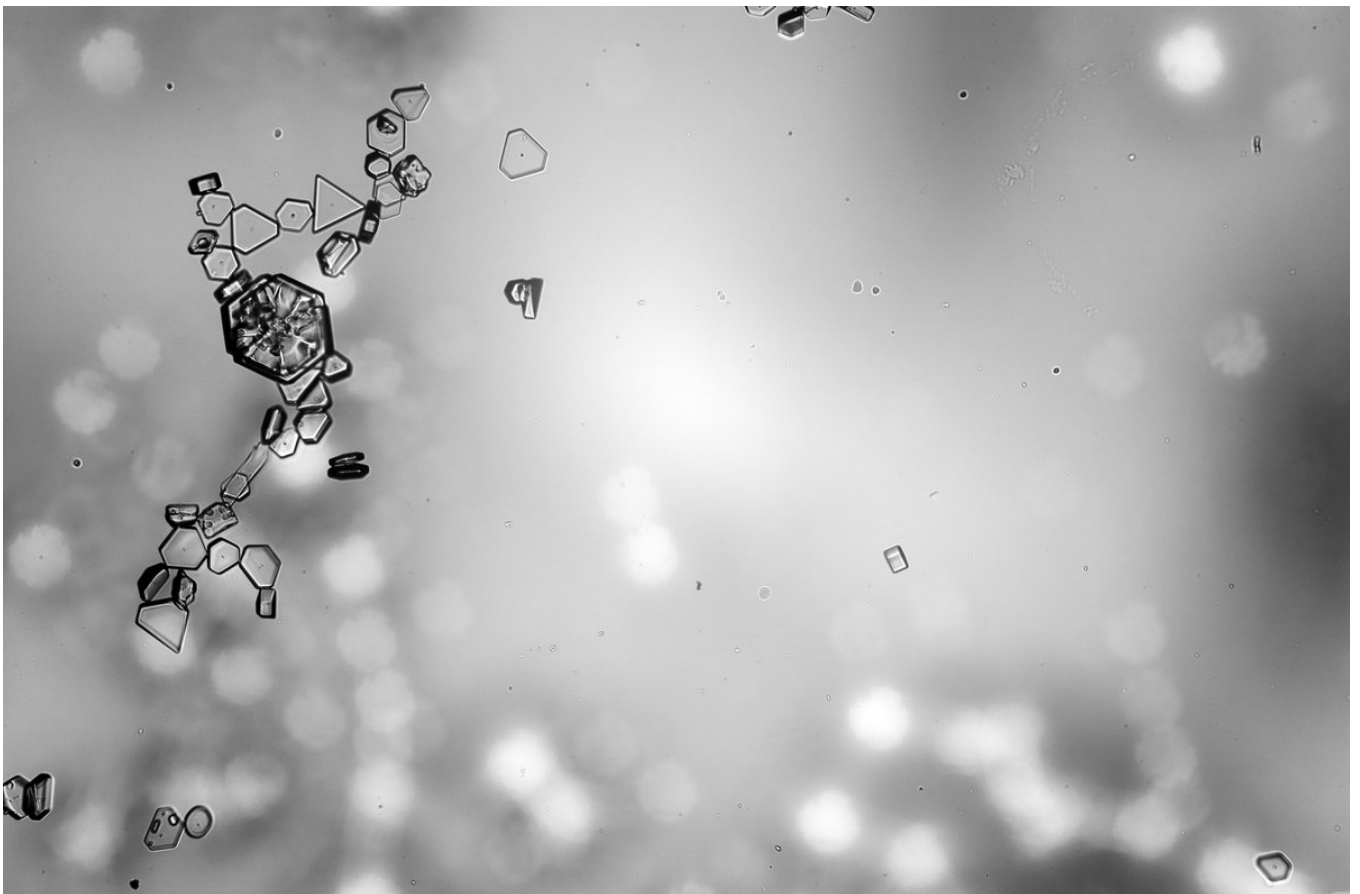
17/18 Feb 2026, Rovaniemi. Crystals of the third swarm.

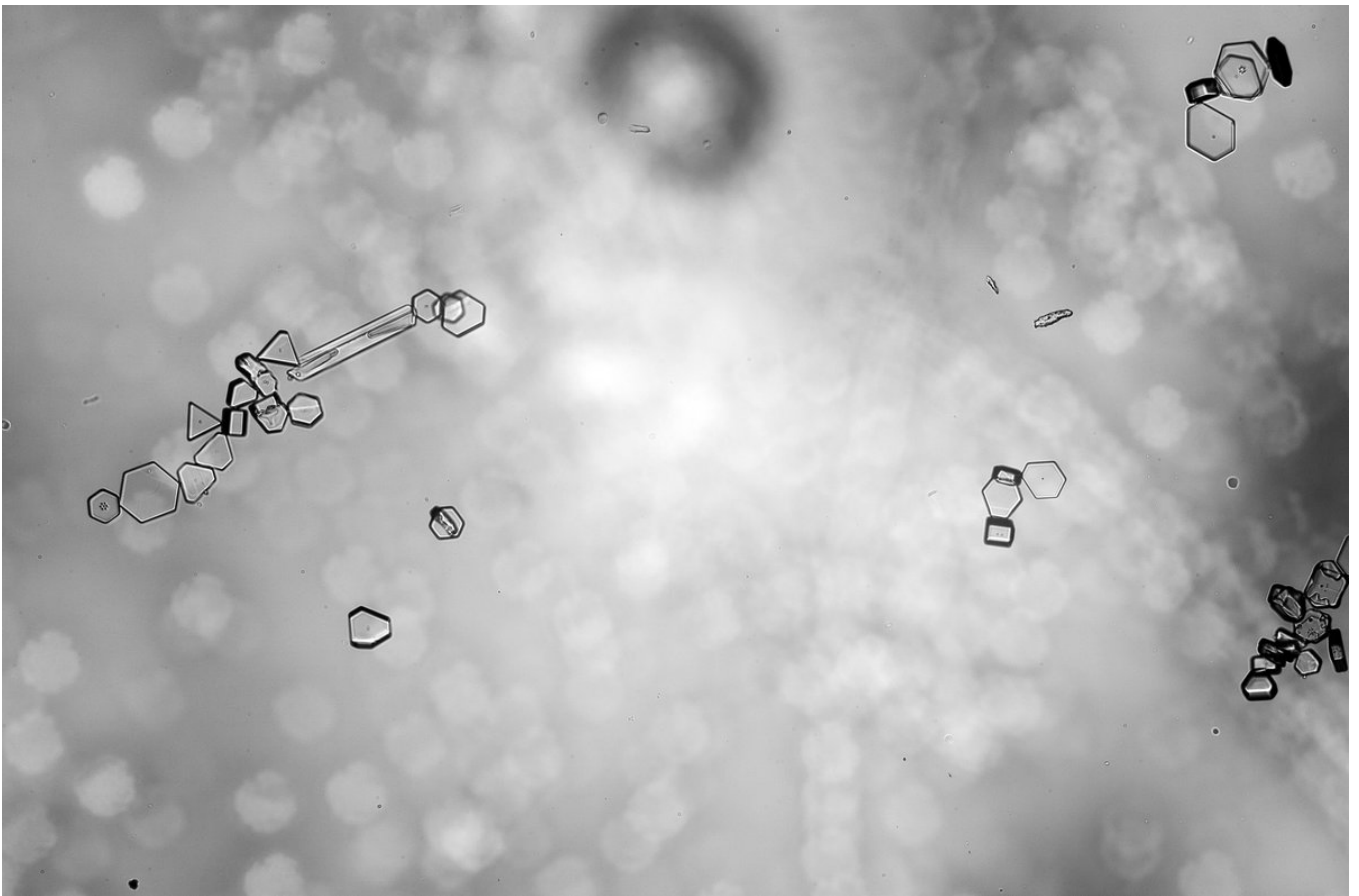
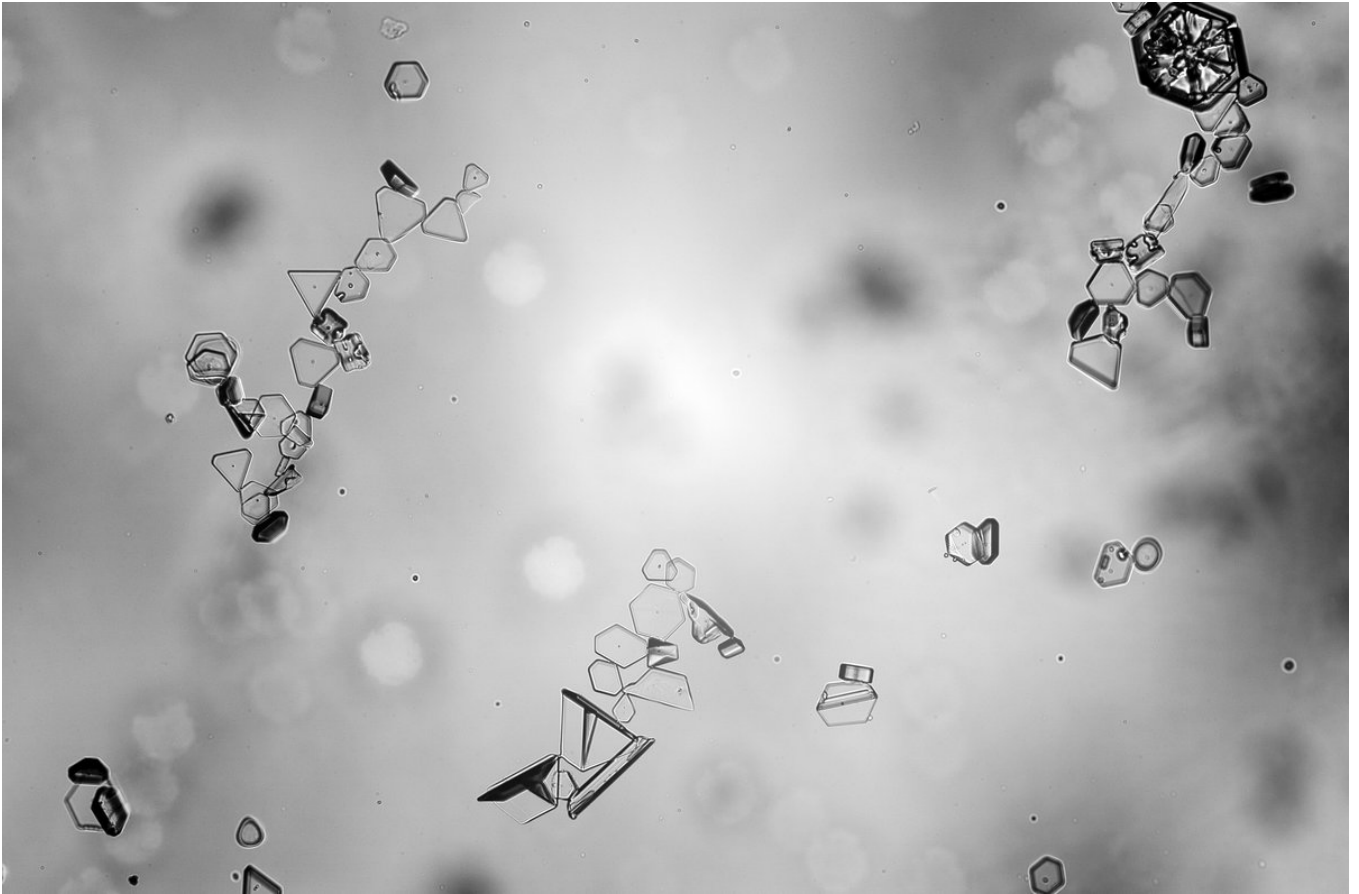




16/21 - 2026-02-22 11:17 UTC

17/18 Feb 2026, Rovaniemi. Crystals of the third swarm.

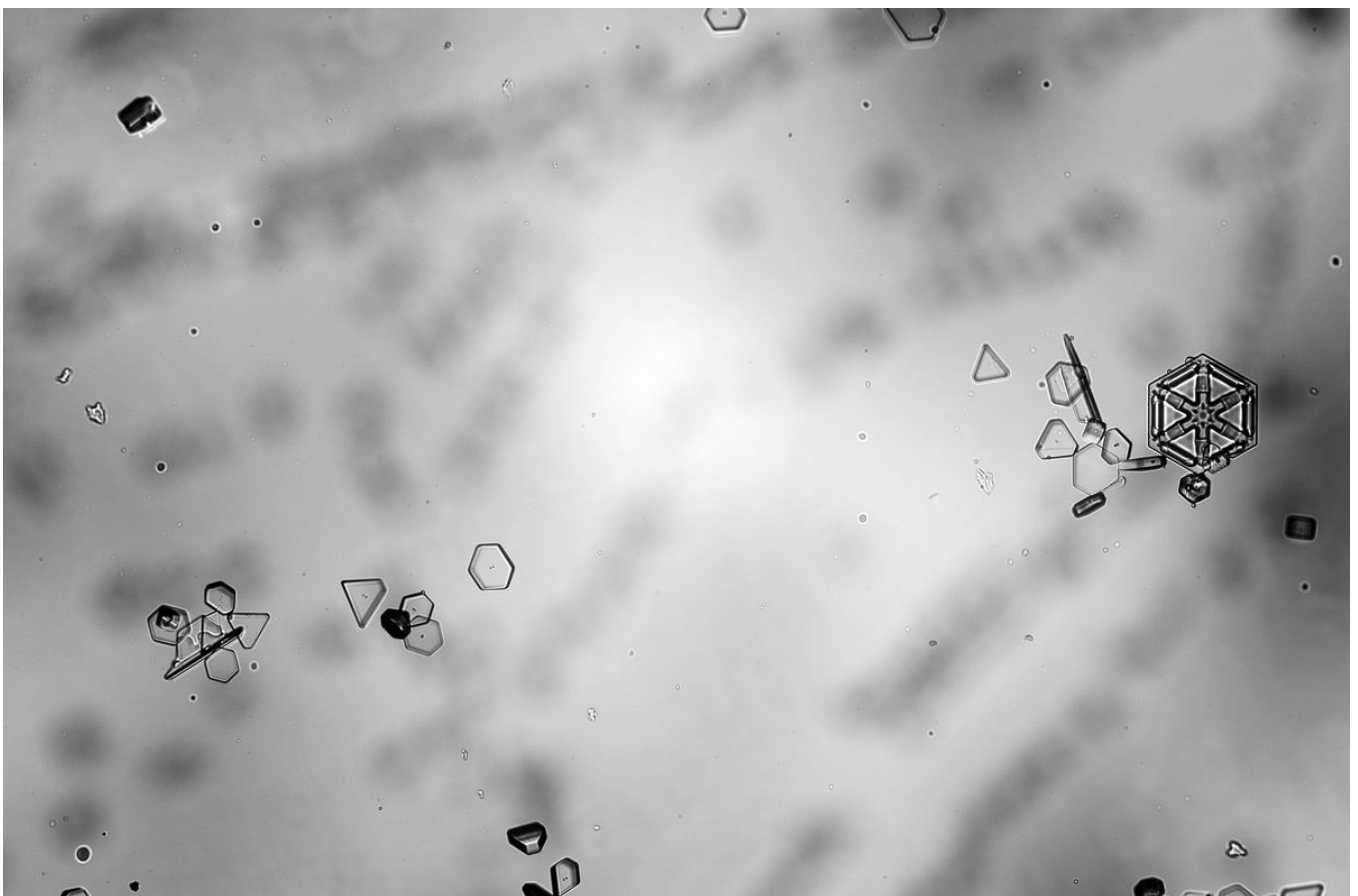


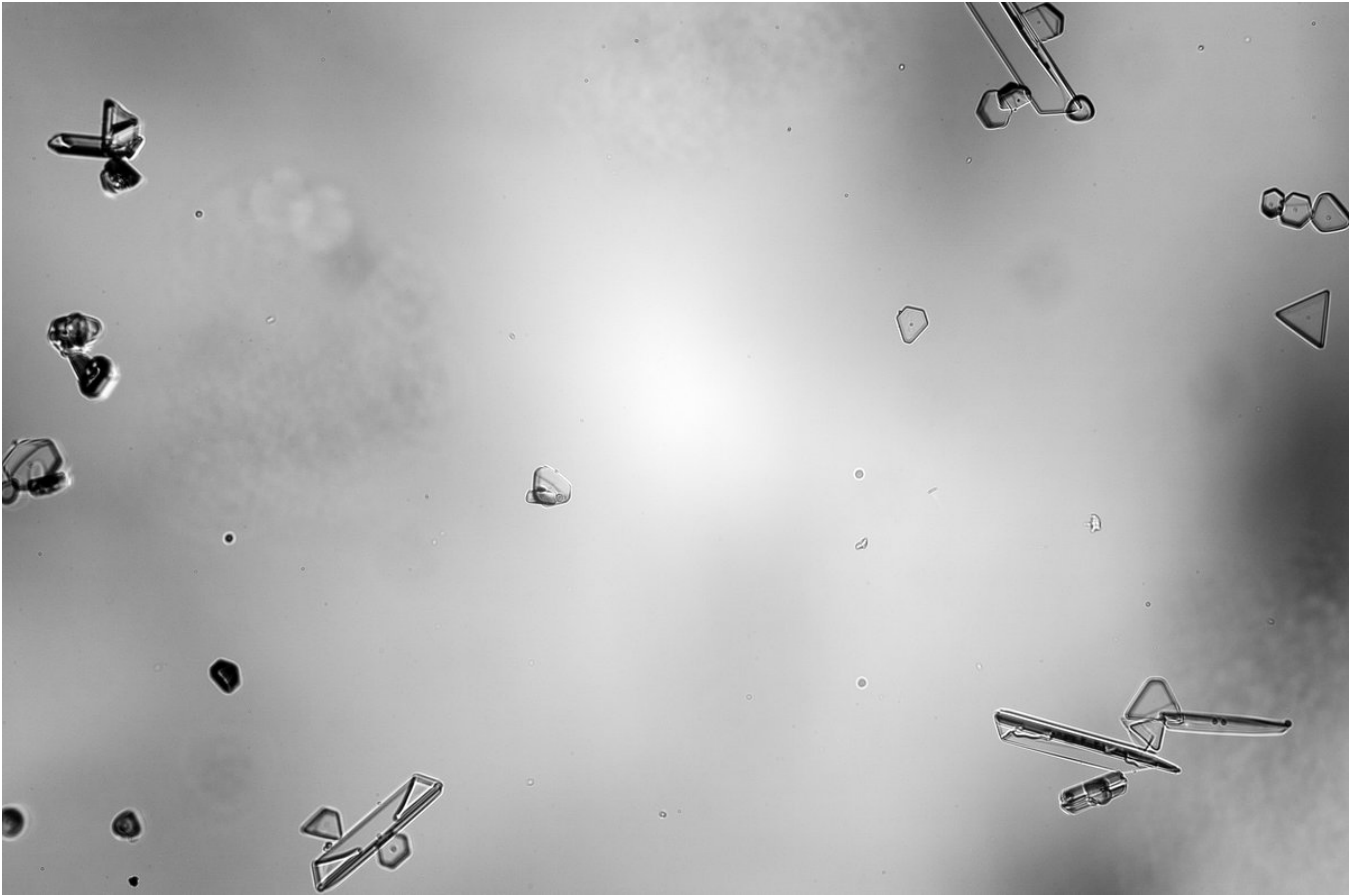
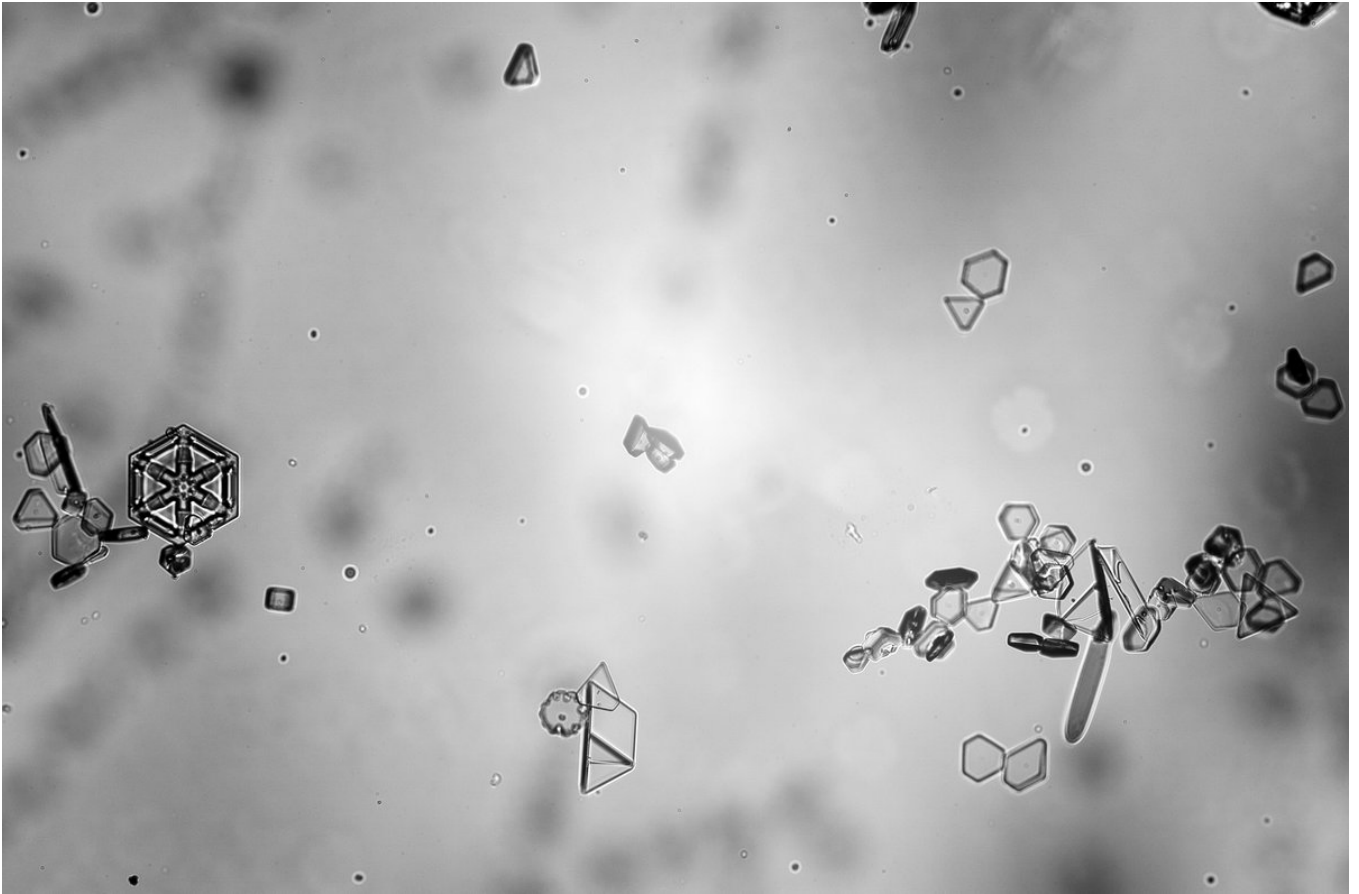


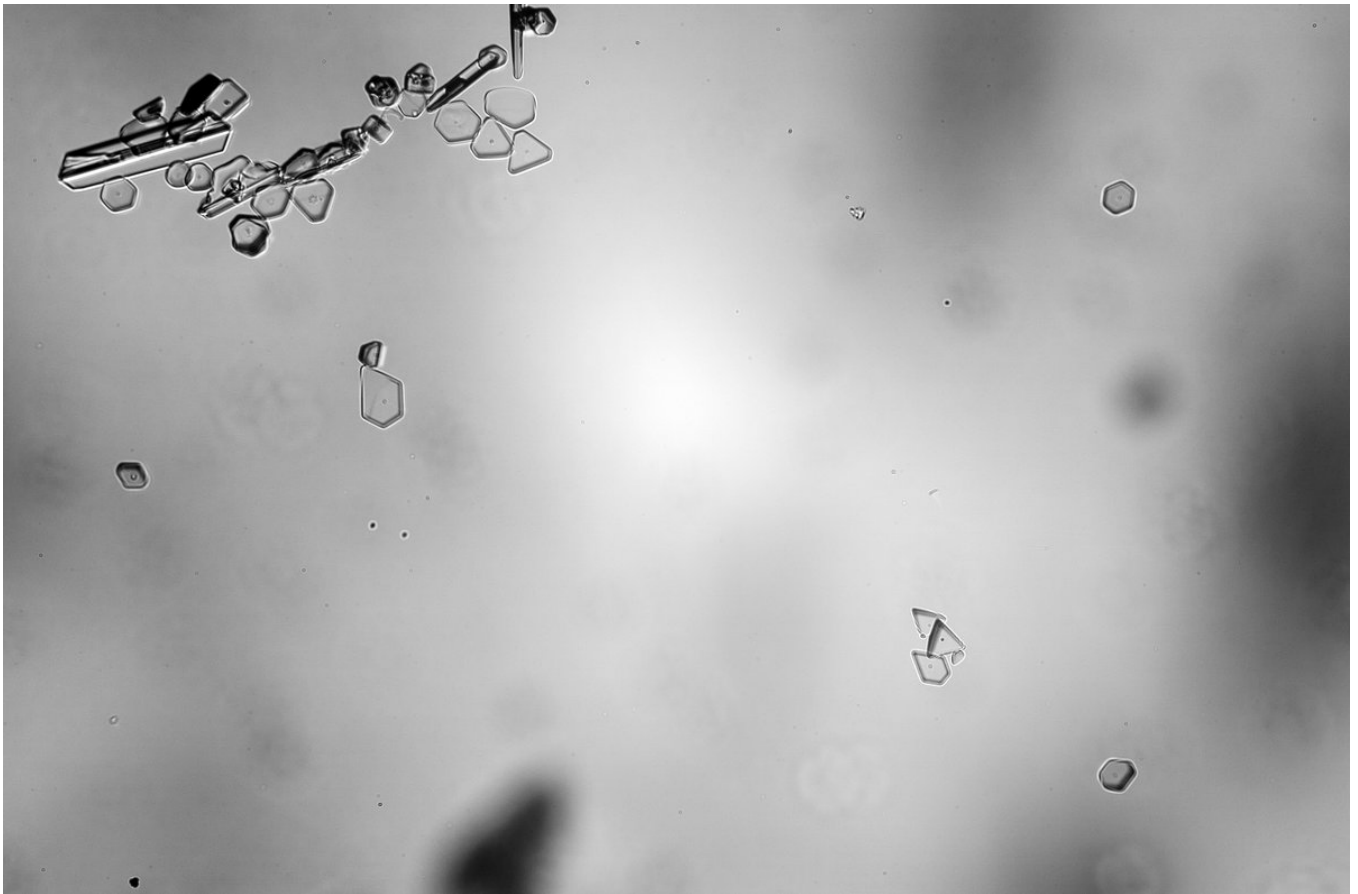


17/21 - 2026-02-22 11:18 UTC

17/18 Feb 2026, Rovaniemi. Crystals of the third swarm.

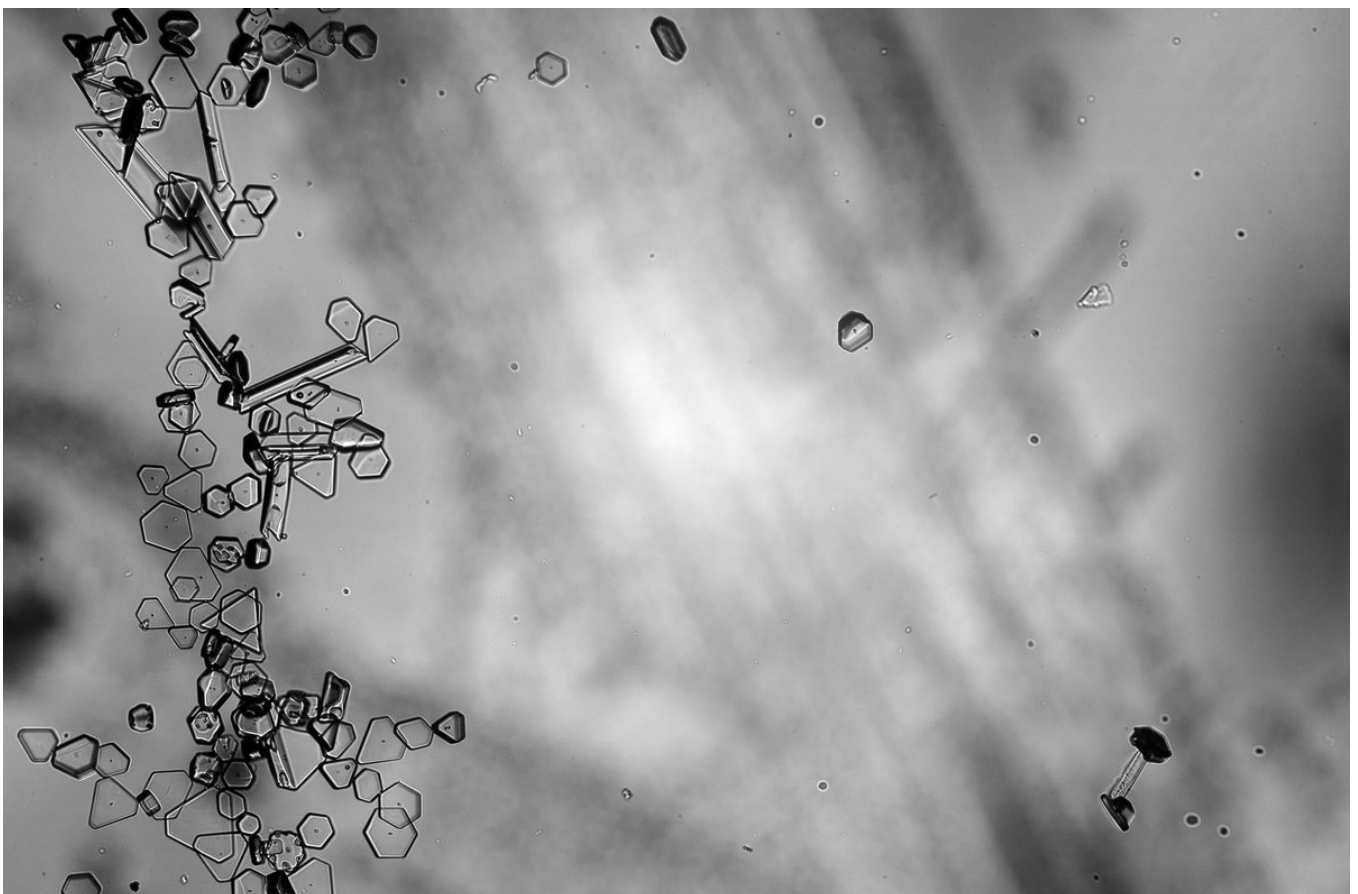


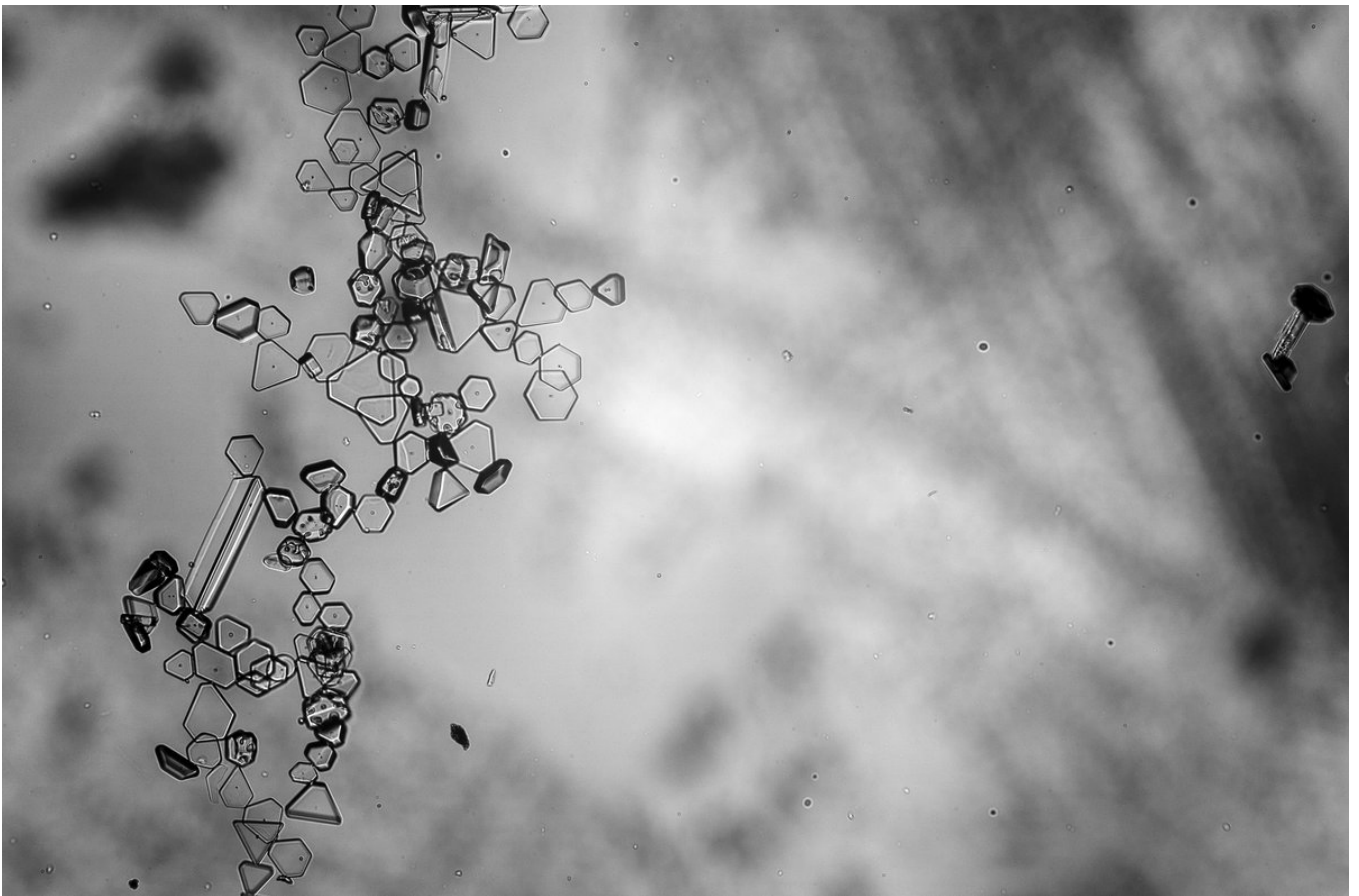
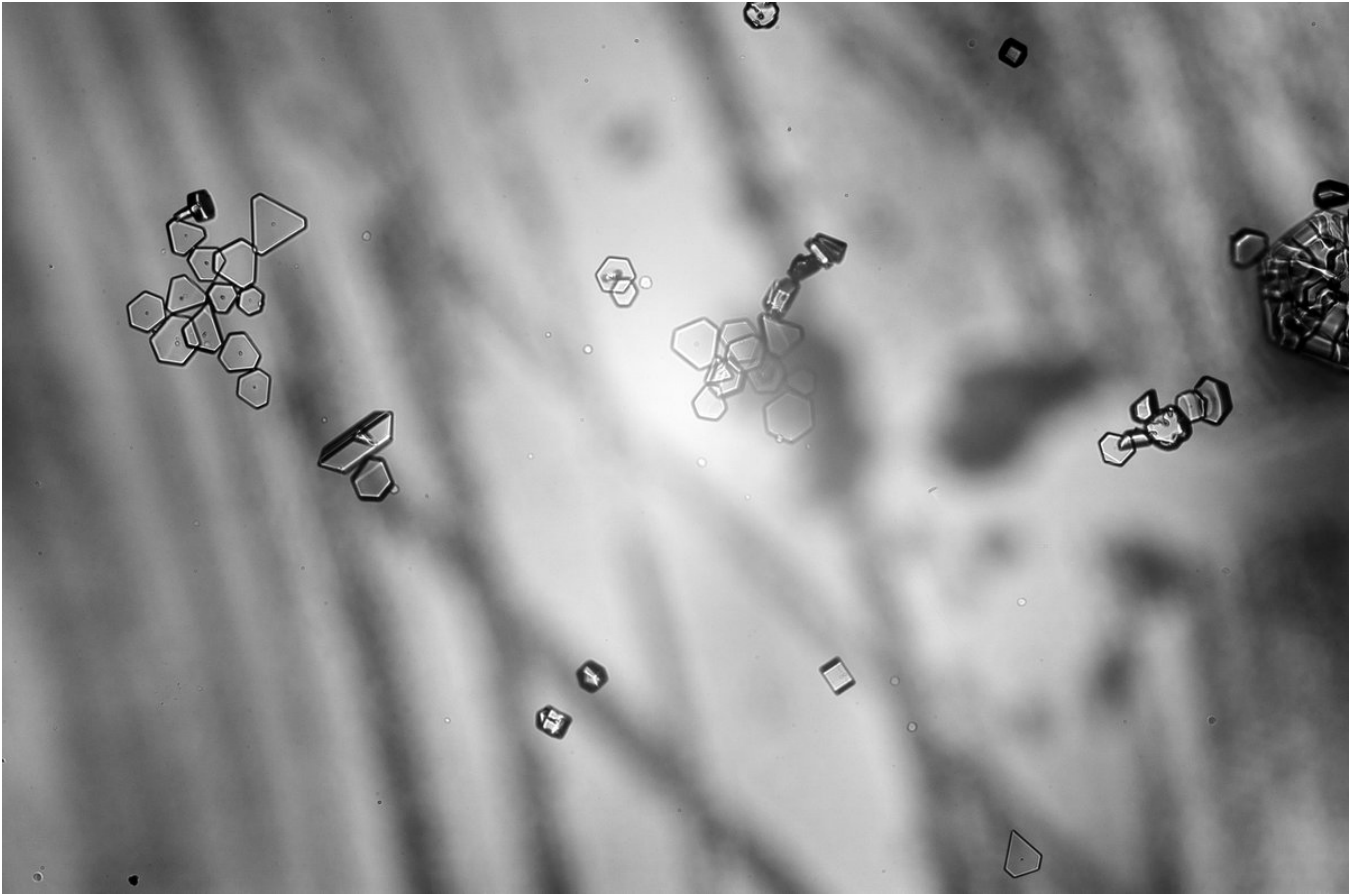


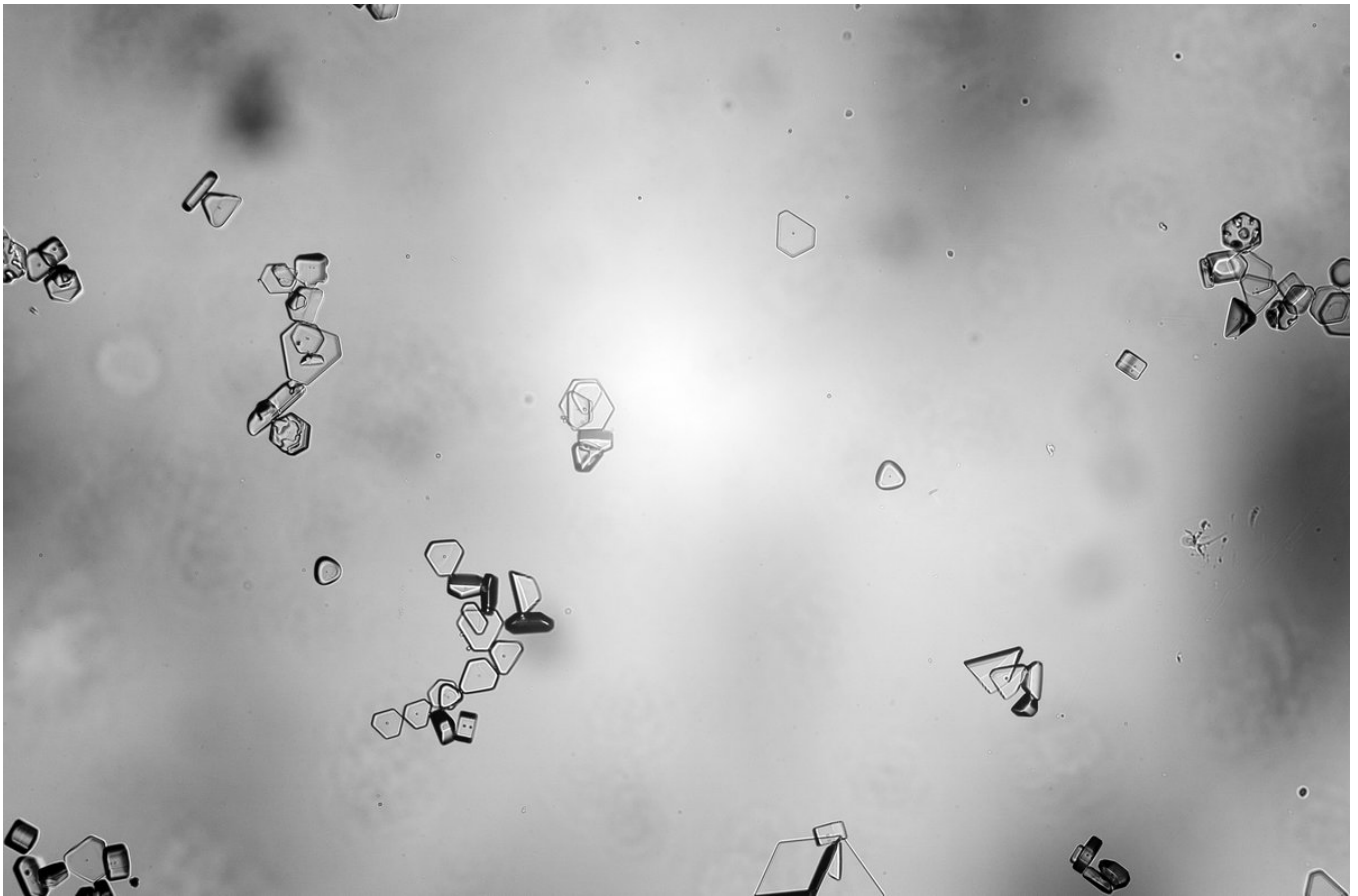


18/21 - 2026-02-22 11:18 UTC

17/18 Feb 2026, Rovaniemi. Crystals of the third swarm.

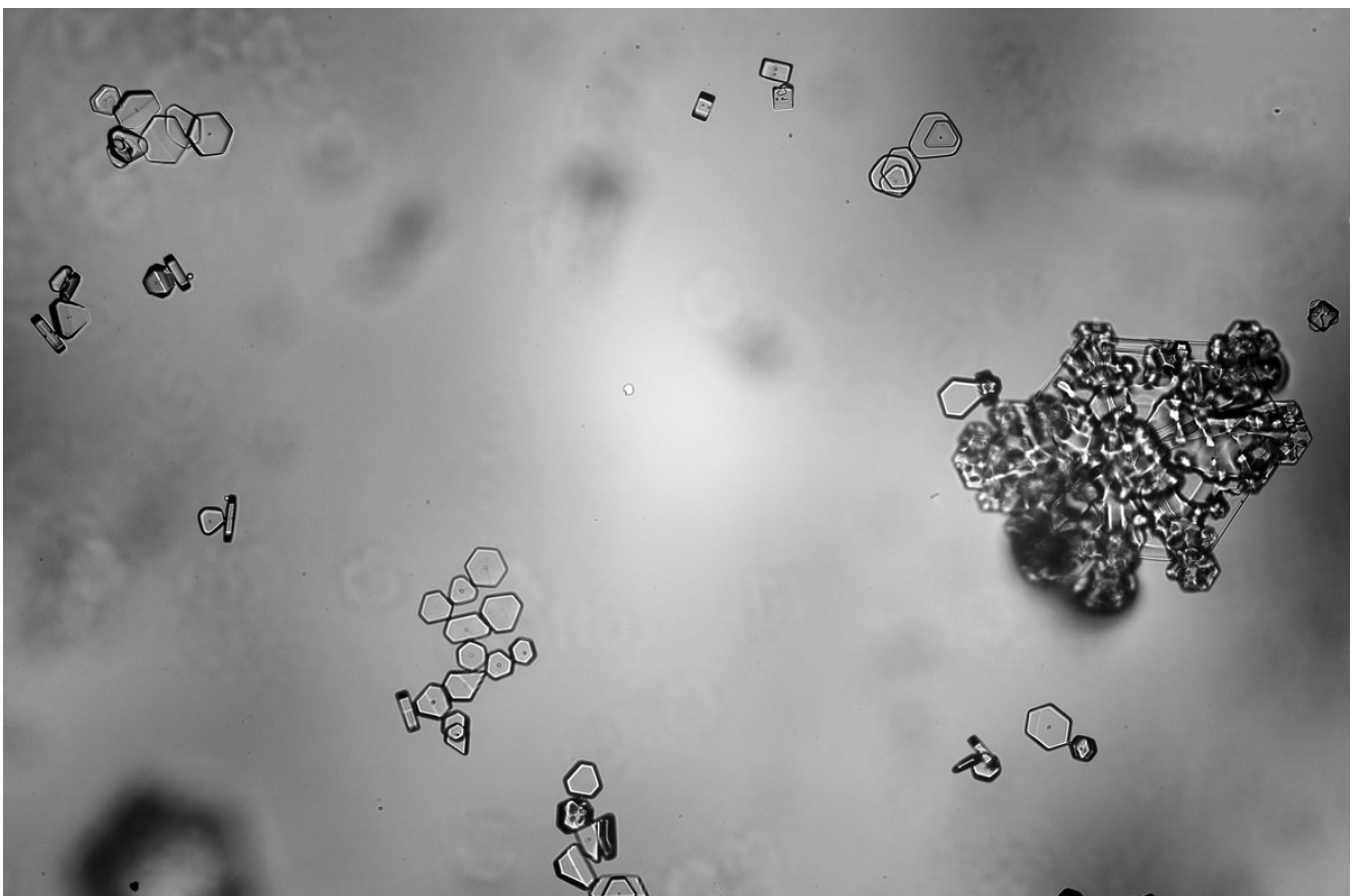


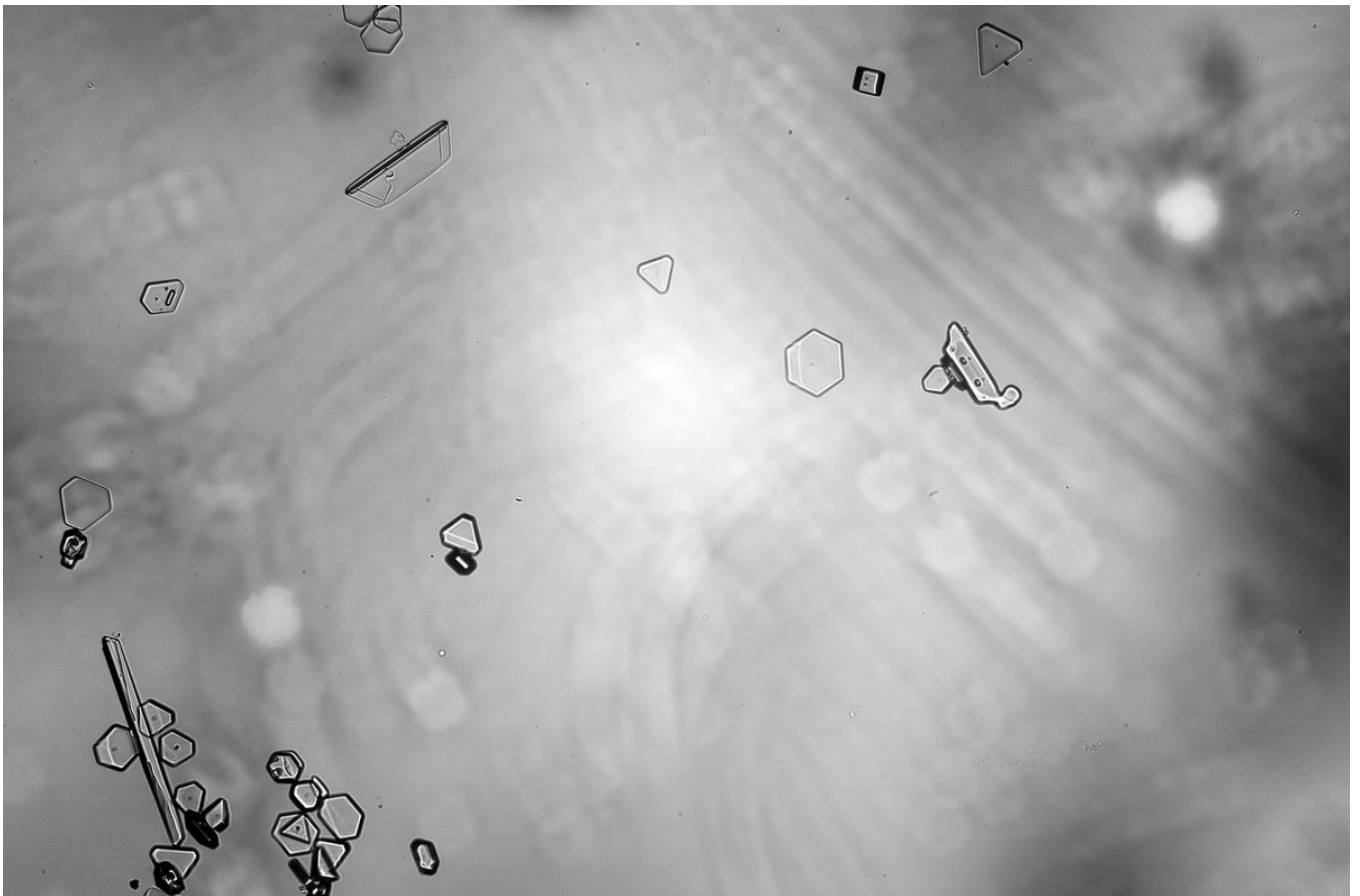
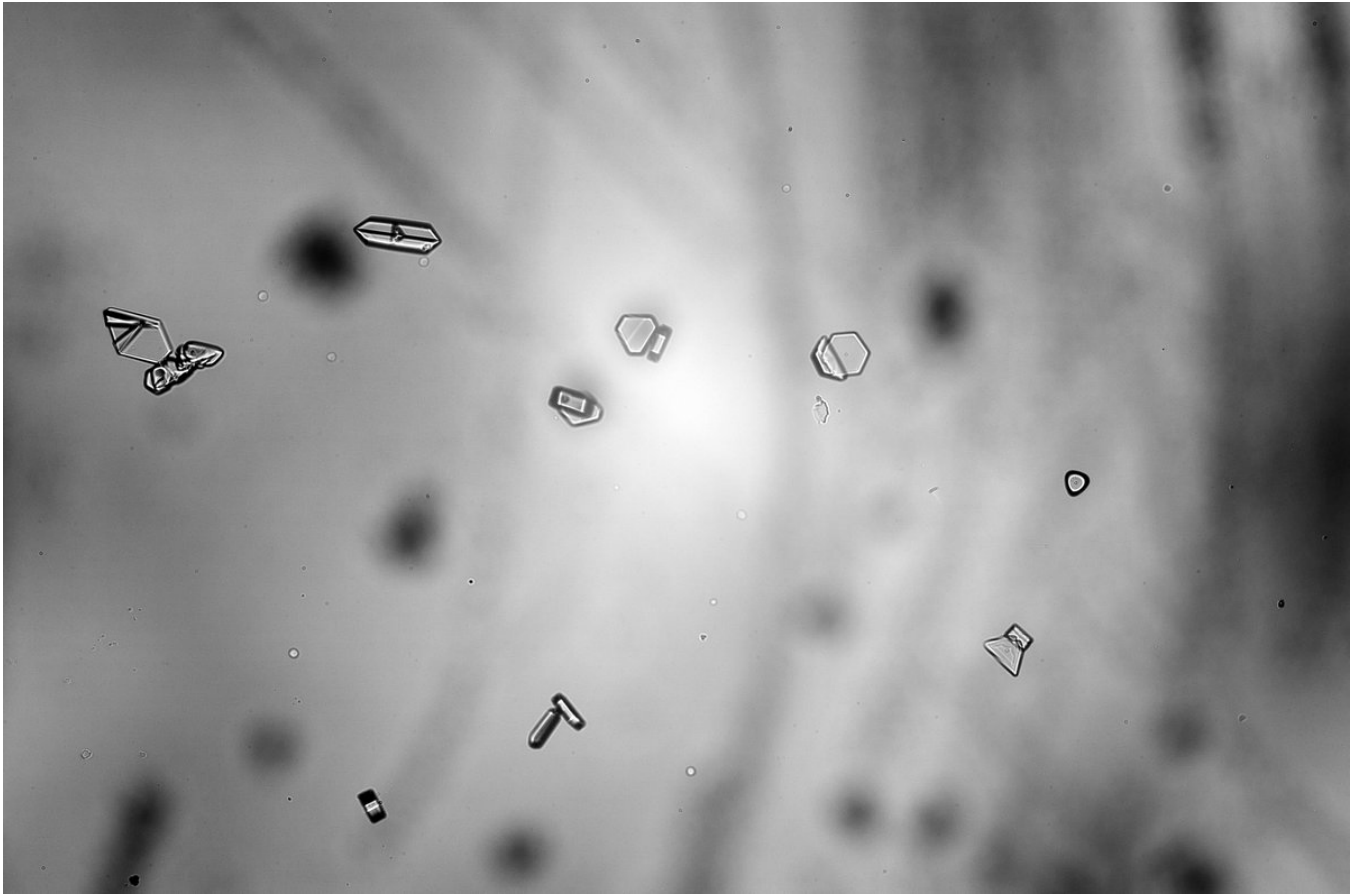


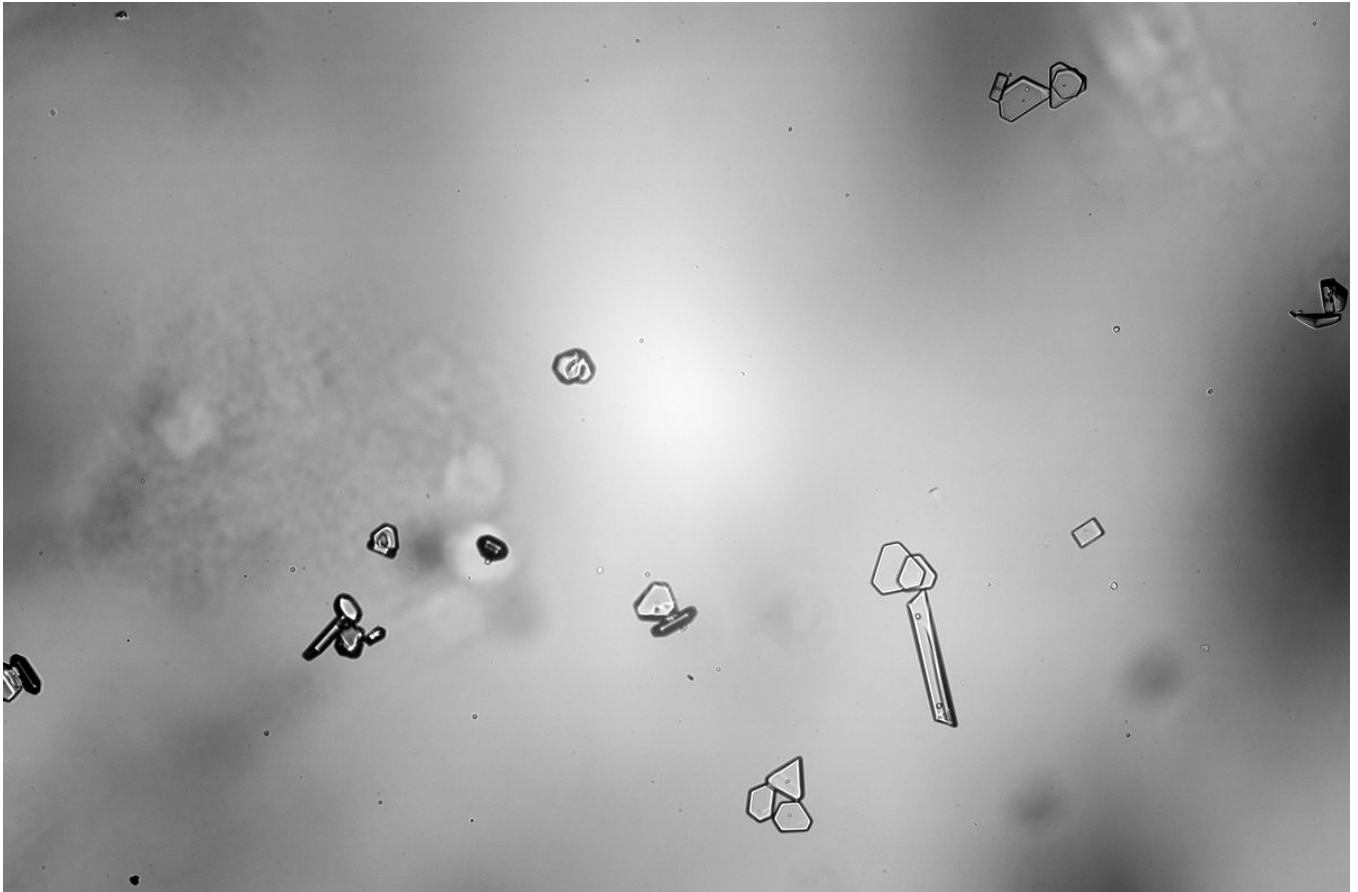


19/21 - 2026-02-22 11:19 UTC

17/18 Feb 2026, Rovaniemi. Crystals of the third swarm.







20/21 - 2026-02-22 13:13 UTC

17/18 Feb 2026, Rovaniemi. I should correct that I actually photographed the second swarm, as is evident from the fifth post in this thread. But I didn't not get the crystals. I was procrastinating and when I finally activated, the swarm was already waning.

21/21 - 2026-02-22 13:36 UTC

17/18 Feb 2026, Rovaniemi. The fading of the 3rd swarm, six successive images. Strikes me how resilient the helic arc is. Parry is good as gone but helic arc hangs on.

